Department of Treasuries
Government of Kerala

REQUEST FOR PROPOSAL

For Selection of Service Provider for Supply, Installation and Maintenance of Software Defined Wide Area Network (SD WAN) Solution

Tender Reference Number: TRY-4902/2022-A4 Dated: 25/03/2025

The Directorate of Treasuries, Pattom Palace, P. O.
Thiruvananthapuram, Kerala -695004

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Disclaimer

Response to Request for Proposal is invited on behalf of **The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala** from the Bidders for Supply, Installation, and Maintenance of Software Defined - Wide Area Network (SD-WAN) Solution and Network Monitoring cum Helpdesk Platform.

The information contained in this RFP document, or any information provided subsequently to Bidder(s) whether verbally or in documentary form by or on behalf of Department of Treasuries-Kerala**,** is provided to the Bidder(s) on the terms and conditions set out in this RFP document and all other terms and conditions subject to which such information is provided. This RFP is neither an agreement nor an offer and is only an invitation by Department of Treasuries-Kerala to the interested parties for submission of bids to The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

The purpose of this RFP is to provide the Bidder(s) with information to assist the formulation of their proposals. While effort has been made to include all information and requirements of Department of Treasuries with respect to the solutions requested, this RFP does not claim to include all the information each bidder may require. Each Bidder should conduct its own investigations and analysis and should check the accuracy and completeness of the information in this RFP and wherever necessary obtain independent advice.

Department of Treasuries-Kerala makes no representation or warranty and shall incur no liability under any law, statute, rules or regulations as to the accuracy or completeness of this RFP. Department of Treasuries-Kerala may in its absolute discretion, but without being under any obligation to do so, update, amend or supplement the information in this RFP.

This is not an offer by Department of Treasuries-Kerala but only an invitation to bid in the selection process initiated by Department of Treasuries-Kerala. No contractual obligation whatsoever shall arise from the RFP process until a formal contract is executed by the duly authorized signatory of Department of Treasuries-Kerala and the Successful Bidder. The Department of Treasuries-Kerala reserves the right to reject any responses or proposals that are non-conforming responses. All proposals submitted will become the exclusive property of Department of Treasuries-Kerala.

Confidentiality

Information contained within this RFP is confidential. No part of the same may be reproduced, in whole or in part, unless specifically required for the Bidder's use in responding to this RFP. Unauthorized disclosure of the RFP or its contents, or failure to observe other requirements contained in this document may result in the Bidder's disqualification from further consideration.

All proposals, presentations and supporting materials submitted in this RFP will be treated with confidentiality and used for the purpose of selection of Bidder. The Department of Treasuries-Kerala reserves the right to conduct further evaluations and to seek further information from the respondents (including technical and financial capabilities) who respond to this RFP.

This document is the property of The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala. It may not be copied, distributed or recorded on any medium, electronic or otherwise, without written permission thereof. The use of the contents of this document, even by the authorized personnel / agencies for any purpose other than the purpose specified herein, is strictly prohibited and shall amount to copyright violation and thus, shall be punishable under the Indian Law.

1. Introduction, Key Events and Dates

1.1. Introduction

The Department of Treasuries, Kerala (herein after referred as "DOT"), was formed as an independent Department on 01.08.1963 Vide G.O (P) No 465/63/Fin, dated 30.07.1963 with Director of Treasuries as the administrative head. The administrative control of the Department at Government level is with the Finance Department. After the formation of Kerala State in 1956 and till 1963, Treasury functioned as part of Revenue Department. From the 1st of August 1963, the Treasury Department started to function as an independent department.

To support vast infrastructure of DOT, The DOT has the Information Communication Technology (ICT) wing to usher in e-Governance in the department. The wing is involved in development and deployment of DOT ICT initiatives to bring about process automation and process re-engineering in the core functioning of the wings of the Department and to strengthen IT infrastructure landscape which includes Network Security, Servers and Applications, End User Computing in the department. Software development is being carried out by the NIC, Kerala State Unit, in consultation and requests from various wings of the department.

The solutions are developed within the e-Governance framework to aid in decision support and Management.

Directorate of Treasuries has its Headquarters at Thiruvananthapuram, and it is operating across Kerala State. The Department has 3 Regional Deputy Directorates, 23 District Treasuries, and 204 Sub Treasuries.

Objective of SD-WAN Procurement:

DOT presently connected through different network service provider's ISP Links via Router at Branch Offices to connect to State Datacentre for accessing the Department of Treasuries Business Critical Applications.

The objective of this project is to upgrade the WAN Infrastructure with modern SD-WAN Solution to enable a more agile and scalable network infrastructure.

The SD-WAN implementation aims to:

1. Simplify network management by having centralized control and automation.
2. Enhance application performance through intelligent traffic steering and optimization.
3. Improve network security by integrating security functionalities into the SD-WAN Architecture

Objective of Network Monitoring and Helpdesk Platform Procurement:

DOT presently monitoring the Network Infrastructure and tracking the End User calls manually. Considering the growing IT estate and amount of call volumes, DOT has planned to implement a Network Monitoring cum Helpdesk Platform to automate the monitoring of Network Infrastructure and Helpdesk to automate the call logging with SLA in place.

The Network Monitoring cum Helpdesk Platform implementation aims to:

1. Establish a centralized Monitoring for Network Infrastructure.
2. Enable compliance tracking and generate reports for Day-To-Day IT issues with the help of Helpdesk.

The SDWAN Solution and Network Monitoring cum Helpdesk Platform project initiative will transform the current manual approach into an efficient and automated system.

This RFP document provides information to prospective Bidders about Project Scope and information on the overall bid process.

The Bidders are requested to read all instructions and go through the forms, terms, requirements and other information in the RFP carefully. Submission of the bid shall be deemed to have been done after careful study

and examination of the RFP with full understanding of its implications. The qualified Bidder can submit their bid as per tender schedule.

1.2. Invitation for Tenders offers

- The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala invites Tender, in two bid system (Technical bid and Commercial bid), from Bidders for Supply, Installation and Maintenance of On-Premises SD-WAN Solution and On-Premises Network Monitoring cum Helpdesk Platform.
- DOT reserves the right to cancel this RFP at any stage of tendering process.
- DOT reserves the right to alter the scope of work at any stage with suitable adjustment.

1.3. Executive Summary

The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala
Here after referred to as the DOT/Owner/End User, reserve its right to accept or reject any design proposal without assigning any reason; its decision will be final.

1.4. Key Events and Dates

The Key Events and Dates are as per the GEM Tender.
Note:

- DOT reserves the right to cancel the tender process at any stage during the tender process time.
- This RFP should be read in conjunction with the Annexures (Annexure No.1 to 14).

1.5. Pre-bid Queries

Bidder’s Request for Clarification to be submitted via GEM Portal.

2. Prequalification Criteria& Evaluation Process

2.1. Prequalification Criteria of Bidder

The Bidders should submit the documentary evidence for fulfilling the required Eligibility Criteria. Documents to be submitted along with the technical and commercial bid of tender document. If a bidder fails to submit the documentary proof for eligibility on time, their bid will be summarily rejected.

Note: Bidders to upload the relevant documents only as mentioned in the RFP. Please make sure unnecessary/non-relevant documents are not uploaded. The upload of unnecessary documents may lead to disqualification.

Sl. No	Prerequisites	Documents to be submitted by Bidder for the prerequisites
1	Bidder should be a company registered in India under the Companies Act	(Copy of the Certificate shall be shared). The Bidder should have been in the similar business for a period not less than **FIVE years** as on tender publication date. (Supporting documents should be enclosed). • Certificate of Registration issued by Registrar of Companies. • GST & PAN Registration Certificate. • Certificate from authorized signatory/ Company Secretary of the bidder indicating that they are in IT Solution for last 5years
2	The bidder should be either an Original Equipment Manufacturer (OEM) or Authorized Dealer / Distributer of the OEM of the offered products	A valid certificate **of Manufacturers Authorization Form (MAF)** should be enclosed. The MAF format is mentioned in **Annexure5.** The bidder / vendor should be submitting the MAF in the same format without modifying the content / sentences.
3	Bidder Turnover	Turnover of the bidder for last 3 individual years (2021-22, 2022-23, 2023-24) should be **Rs.10 Crore or above**. (Relevant pages of the Audited Financial Statement shall be uploaded). Audit Balance Sheets of last three FY viz. FY 2021-22, FY 2022-23 & FY 2023-24. **CA Certificate** indicating the **turnover for the last 3 years**, and **Positive Net worth** as on **31 March 2024**.
4	Project Experience	Bidder should have experience and expertise of having successfully completed any IT Network projects of the following value, in addition to 1 SD-WAN Purchase Order, Any 1 Purchase Order of value 7 Crores or above, (OR) 2 Purchase orders of value 4 Crores or above (OR) 3 Purchase orders of value 3 Crores or above. ***Note: All Purchase Orders must be from Government Organisations in India.***
5	Declaration of non-Blacklisting	The Bidder should have not been blacklisted at the time of submission of the bid by any regulatory body / statutory body/ any Government department/ PSU/PSE/ Private sector
6	Declaration on Proposed	The bidder is required to quote for the complete BOQ. Partial quotes

	BOQ	are liable to be rejected.
7	Confirmation on Delivery of the Hardware and Software	The bidder should be ready with all the deliverables including all Hardware and Software items within 60 working days from the date of Purchase Order.
8	Bidder's Local Office and Escalation Matrix	Bidder should have Support Office at Kerala and should submit the escalation matrix.

2.2. Evaluation Methodology QCBS

2.2.1 Bids shall be evaluated both in terms of '**Quality'** as well as '**Quoted Price**' i.e. Quality & Cost Based Selection (QCBS) methodology. **The weightage for the 'Quality' is 70 (Seventy) and the weightage for the 'Quoted' price is 30 (Thirty).**

2.2.2 The marks allocated against various sub-sections under 'Quality' of Bid shall be as hereunder:

Sl. No	Category	Max. Marks	Description	Marks	Document(s) to be Submitted
1	SDWAN OEM Experience Government Department/BFSI/ PSU/PSE/Private sector	10	SDWAN Experience in India (As on Bid calling Date) a) More than 10 years - 5 Marks b) More than 7.5 years - 3 Marks c) More than 5 years - 2 Marks	5	PO copy with Date clearly mentioned
2			SDWAN Experience in India a) More than 500 multisite Project - 5 Marks b) More than 250 multisite Project - 3 Marks c) Less than 100 multisite Project - 2 Marks	5	PO copies
3	SD-WAN Technical Specifications (Core functionalities)	35	Capability to integrate different types of connectivity (MPLS/ILL/FTTH/Broadband/SIM/VSAT) into a single, end-to-end IP Framework and IP tunnel that uses policies to manage traffic.	5	Data Sheet and PoC
4			Capability to integrate different types of connectivity for Active- Active Load Balancing without traffic duplication on the connected Links at a site (MPLS/ILL/FTTH/Broadband/SIM/VSAT)	15	Data Sheet and PoC
5			Capability for DNS based Inbound Load Balancing of Multiple divergent media	5	Data Sheet and PoC
6			Capability of Seamless Failover of Voice/Video and Data Traffic less than 20Milliseconds	5	Data Sheet and PoC
7			Capability for switching traffic to new DHCP IP's from Service Providers for multiple DHCP based Links at sites	5	Data Sheet and PoC
8	SD-WAN Technical Specifications (Security)	15	Capability of inbuilt Firewall and IPS in DC/DR/Branch Devices	5	Data Sheet and PoC
9			Capability of Selective Encryption of Links at sites	5	Data Sheet and PoC
10			Traffic Encryption capability shall be FIPS 140-2 Compliant	5	Data Sheet and PoC
11	SD-WAN Technical Specifications (Application Awareness)	5	Capability to Prioritise and steer traffic based on Applications	5	Data Sheet and PoC

12	SD-WAN Technical Specifications (WAN Optimization)	10	Built In Advanced WAN Optimisation Capabilities - Caching, Compression, Deduplication of Application traffic	5	Data Sheet and PoC
13			Capability of WAN Optimisation at Byte Level	5	Data Sheet and PoC
14	The proposed Network Monitoring cum Helpdesk Platform OEM Experience	5	Network Monitoring cum Helpdesk Platform Experience in India - (in state/central Government/ PSU) a) Minimum 3 Projects with 10000 Devices - 5 Marks b) 2 Projects with 10000 Devices - 3 Marks c) 1 Project with 10000 Devices - 1 Marks	5	Reference PO copy and completion/ Signoff Documents need to be submitted.
15	The proposed Network Monitoring cum Helpdesk Platform OEM certifications	5	a) The Proposed Network Monitoring and Helpdesk Platform OEM should be OWASP & SANS certified from cert-in empanelled vendor. b) The Proposed Network Monitoring cum Helpdesk Platform OEM should be certified for ITIL V4 for 9 or more practices. c) The proposed Network Monitoring cum Helpdesk Platform OEM should have ISO 270001, ISO 27034, SOC3 and CMMI Level 3	5	Documents need to be submitted for all 3 points
16	Presentation by Technically Qualified OEM's	15	Each SD-WAN and Network Monitoring cum Helpdesk Platform OEM's will be allotted 30 minutes time slot for presenting (ON-LINE) the technical capabilities, features, proposed components, and deployment approach in line with RFP requirements. SD-WAN OEM - 10 Marks Network Monitoring cum Helpdesk Platform OEM -05 Marks	15	Presentation to Technical Committee
	TOTAL	100		100	

2.2.3 The technical evaluation and the commercial evaluation shall have the weightage of **70% and 30%** respectively and this weightage shall be considered for arriving at the successful bidder. The evaluation methodology vis-à-vis the weightages is as under:

2.2.3.1 The vendor needs to achieve a cut – off score of **50 marks** in this evaluation stage **to be qualified for commercial bid opening**. Only those vendors who achieve the specified cut – off scores would be short-listed for Commercial Bid Evaluation.

2.2.3.2 100% compliance to the Technical Specifications requirements is mandatory as mentioned in **Annexure 7 (B), 7(C**), **7(D**) and **7(E**) to qualify in the RFP process.

2.2.3.3 The Technical Proposal will be evaluated for technical suitability and the criteria for evaluation of technical bids as per the Quality Criteria.

2.2.4 Quality Criteria:

2.2.4.1 Qualified Bids' (meeting the minimum Qualifying Marks in Quality Criteria) and conforming to the technical specifications, terms and conditions stipulated in the bidding document and

considered to be responsive after subjecting to Bid Evaluation Criteria shall be considered for further evaluation as per the Evaluation Criteria given below:

2.2.4.1.1 Price Bids shall be evaluated taking into account the Price quoted for all services including applicable GST (CGST & SGST / UTGST or IGST).

2.2.4.1.2 Quoted price must include all liabilities and taxes including statutory liabilities but excluding GST, which shall be quoted separately in the Price Bid format.

2.2.4.1.3 To ascertain the Inter-se-ranking of the bids, the Quality & Cost Based Selection (QCBS) methodology as mentioned below shall be adopted:

2.2.4.1.4 An Evaluated Bid Score (B) will be calculated for each bid, which meets the minimum Qualifying marks of 70 (Seventy-Five) in 'Quality' Evaluation Criteria, using the following formula in order to have a comprehensive assessment of the Bid price and the Quality of each bid:

B = ((C-Low / C) * 100 *X) + ((T/T-high) * 100 *Y)

Where,
C = Evaluated Price of the Bidder
C-Low = the lowest of the evaluated bid prices among the responsive bids
T = the total marks obtained by the bidder against "Quality" Criteria
T-high = the highest marks scored against "Quality" criteria among all bids
X = 0.30 (The weightage for quoted price is 30%)
Y = 0.70 (The weightage for "Quality" is 70%)

Note: The Evaluated bid score (B) shall be considered up to two decimal places.

2.2.4.1.5 Contract shall be awarded to the bidder with the highest Evaluated Bid Score (B).

2.2.4.1.6 In the event of two or more bids having the same highest Evaluated Bid Score (B), the bid scoring the highest mark against 'Quality' criteria will be recommended for award of contract. Even if there is a tie, PoC & Technical Presentation shall be conducted to decide the preferred Bidder for the awarding the Purchase Order.

2.2.4.1.7 To ascertain the inter-se-ranking, the comparison of the responsive bids will be made subject to loading for any deviation.

Note: POC for all the Quality criteria points must be performed by the H1 bidder in order to get the Purchase Order for the project. If H1 Bidder fails to successfully perform the PoC, the opportunity will be awarded to H2 bidder.

3. Project Overview

3.1. Bidder and OEM responsibilities

- The Bidder is required to Supply, Install and Maintain the SD-WAN Solution and Network Monitoring cum Helpdesk Platform.
- The OEM and successful Bidder must study the existing setup of DOT and implement the proposed SD-WAN Solution and Network Monitoring cum Helpdesk Platform as per the industry best practices.
- The OEM and Bidder needs to understand the requirement of DOT and prepare detailed implementation plan for SD-WAN Solution and Network Monitoring cum Helpdesk Platform. On approval of the same by DOT, implementation needs to be carried out. Detailed Architecture, design, traffic flow and policies should be documented. Deployment will start only after acceptance by DOT.
- The Design, Implementation and Configuration of the SD-WAN Solution and Network Monitoring cum Helpdesk Platform must be taken care by respective OEM along with Bidder.
- The OEMs of SD-WAN Solution and Network Monitoring cum Helpdesk Platform along with Bidder to provide Knowledge Transfer to DOT's administrators. KT shall cover all HLD/LLD details, configurations and administration activities.
- The DOT will provide the required infrastructure (Hardware/Software, OS & Virtualization platform, DB) to deploy the proposed Network Monitoring cum Helpdesk Platform. The bidder shall provide all the required Hardware and Software details.
- The Bidder must deploy two Onsite Technical Engineers to manage proposed SD-WAN Solution cum Network Infrastructure and Network Monitoring cum Helpdesk Platform. Please refer the job responsibilities of the Technical Engineers which are mentioned in the **Section 5.12.1 and 5.12.2.**
- The Bidder should have back-to-back support with respective OEM's during the entire contract period for necessary support. The Bidder should have necessary agreement with the OEMs for all the required support for entire project period.
- The Bidder must ensure that DOT gets all necessary support from the OEM Technical Assistance Centre team to address technical issues for timely resolution. OEM TAC provide the support throughout the period of the contract via remote, Web, Email and Telephonic.
- The Bidder and OEM should provide escalation matrix within their organization.
- The Bidder to make arrangements with OEM or OEM authorized Training agency to provide on-premises training on SD-WAN Solution and Network Monitoring cum Helpdesk Platform for nominated members of DOT.
- The OEM and Bidder should coordinate with existing System and Network Integrators/Telecom Service Providers, other vendors and DOT's Officials during Design, Installation, and Configuration.
- The OEM to ensure the proposed solutions must not be declared End of Life (EoL) for a period of 5 Years and End of Support (EoS) for a period of 7 years from the date of commissioning of the solution. Declaration to be provided along with Bid. In case, OEM declares their product's end of life during the contract period, OEM should provide upgraded version of the products without any additional cost to the DOT. The respective SD-WAN Solution OEM and Network Monitoring cum Helpdesk Platform OEM to declare the same using **Annexure 12 - End of Sale/End of Support.**
- The OEM and Bidder is required to propose SD-WAN Solution and Network Monitoring cum Helpdesk Platform as mentioned in the RFP, with no deviation or conditions. In case the SD-WAN Solution and Network Monitoring cum Helpdesk Platform is found to be undersized and/or not meeting the DOT's requirements, the bidder will be responsible to enhance the sizing and deliverables as per DOT's requirement stated in this RFP at no additional cost, during the contract period. The proposed solution should meet all the technical criteria mentioned in the RFP.

- The OEM would be responsible for replacing and upgrading the out-of-support, out-of-service, end-of-life, undersized at no extra cost to the DOT during the entire contact period. Replacement to be done before due date of the product/service.
- The proposed solution to be deployed under scope of this RFP, should be complete in all respects. The DOT will not accept any plea of the Bidder OR OEM later for omission of services on the pretext that the same was not explicitly mentioned in the RFP.
- The Bidders and OEMs are advised to submit their bids taking full notice of all the pre-qualification criteria if any, technical specifications, terms and conditions. DOT will not be responsible for any assumptions by the Bidder or OEM w.r.t sizing and deliverables.

3.2. Scope of Work

3.2.1 SD-WAN

DOT is seeking to implement the Software-Defined Wide Area Network (SD-WAN) solution to upgrade existing WAN Infrastructure with a solution that leverages software-defined technology to provide increased flexibility, better application performance, and optimized links utilization.

Note: In line with the requirement, The DOT has envisaged SD-WAN Solution Components, Reference Architecture, Deployment Approach and Project Scope which are defined in detail under **Section 3.2.1.1, 3.2.1.2 and 3.2.1.3**.

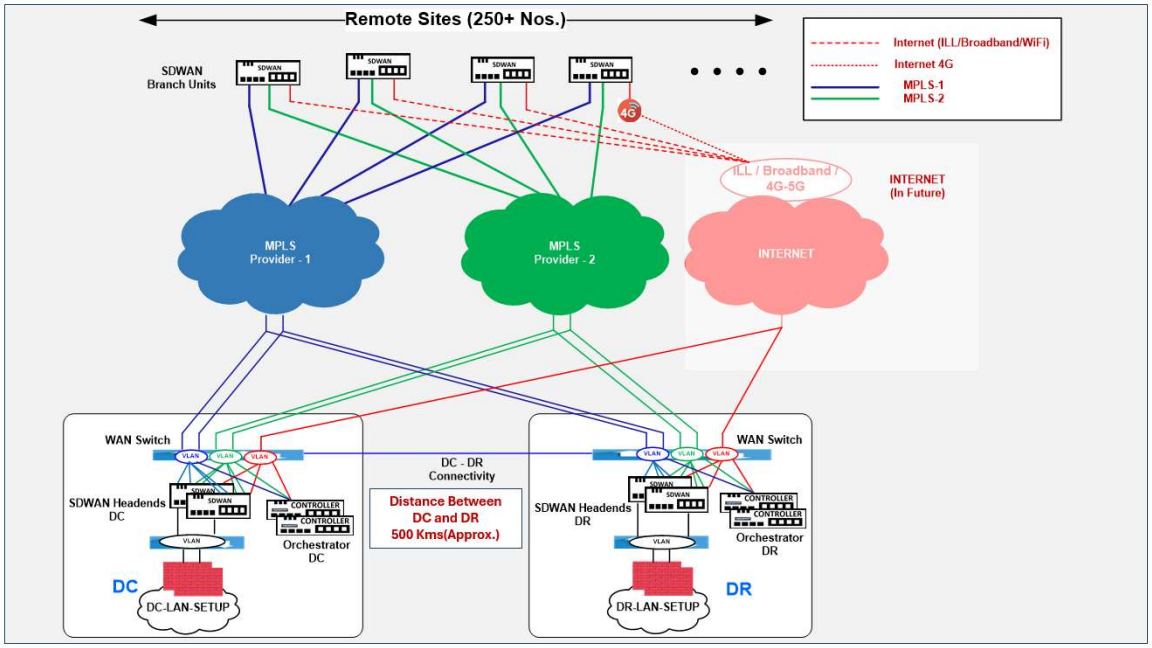
The SD-WAN Solution Components, Reference Architecture, Deployment Approach and Project Scope are not limited to the details provided below and may change from OEM to OEM. DOT reserves the right to ask Bidder along with OEM to conduct POC and demonstrate all or some of the points mentioned in the technical specification section.

3.2.1.1 SD-WAN Solution Components

- **Central Management solution/Controller/Orchestrator:**
  Device installed at DC and DR henceforth called as Central Management solution/Controller/Orchestrator shall have the following capabilities and will be installed in HA at DC and DR.
  a) Device authentication and Management
  b) Pushing configuration to branch SDWAN device and head-end devices.
  c) Templates creation, storage and management.
  d) Route manipulation and configuration management.
  e) Defining the policies and application aware routing.
  f) Monitoring head-end devices and Branch Devices.
  g) The failure of the central Management solution/Controller/Orchestrator shall not impact the DOT Branch and DC and DR data plane and DOT applications traffic.
- **Head-end Devices at DC and DR**:
  The device(s) proposed to be installed at DC and DR for establishing IPSEC tunnel with branch devices to forward data traffic.
- **Branch Devices (Type-1 and Type-2)**:
  The proposed devices being installed under SDWAN solution at DOT's branch offices.
- **Additional Components**
  a. Professional Display Monitor
  75" Professional Display for monitoring the configured SD-WAN Solutions Dashboard in HQ
  b. Desktop
  Desktop with latest High-End configuration for the On-Site Technical Engineer

NOTE: The above are the components envisaged by DOT, the Bidder & OEM must consider & factor all necessary components to accomplish the requirements of this RFP completely.

3.2.1.2 Reference SD-WAN Architecture



Disclaimer: Reference Architecture Representation

The reference Architecture provided above is intended solely for representation purposes. It serves as a conceptual illustration to demonstrate the possible deployment and integration of the proposed solution within a typical environment.

This representation is not to be interpreted as the actual topology or design required for the implementation of the SD-WAN Solution in this RFP. Final topology, configurations, and design will be tailored and determined based on the specific requirements, constraints, and approvals by DOT.

The actual design, plan and implementation will be discussed with OEM and Bidder during the solution design phase.

3.2.1.3 Deployment Approach and Project Scope for SD-WAN

The selected Bidder and OEM will collaborate with DOT team to design an SD-WAN Architecture that aligns with DOT's business objectives and technical requirements. This includes but not limited to the following.

1. **Network Assessment and Design**
   - Conduct a detailed assessment of the current WAN infrastructure and network requirements.
   - Design the SD-WAN Architecture tailored to the DOT's network and security requirements.
   - Create a roadmap for phased implementation, including a migration plan for each site.
2. **Installation of SD-WAN Central Management solution/Controller/Orchestrator, SD-WAN Head End Devices and SD-WAN Branch Devices**
   - Install SD-WAN Central Management solution/Controller/Orchestrator in DC and DR in HA Mode
   - Install SD-WAN Head End Devices in DC and DR in HA Mode
   - Install SD-WAN Branch devices in Standalone Mode
   - Ensure the proposed SD-WAN Solution meets the Technical Specifications mentioned in **Annexure7 (B)** while designing the solution for implementation.
   - Coordinate with the DOT's team for SD-WAN devices connectivity at Branch Offices
   - Establish secure connectivity between Data Centre and Branches

- Define &Implement policies for traffic encryption, load balancing, traffic prioritization & failover
- Configure QoS policies to prioritize critical applications and manage bandwidth effectively.
- Set up failover and load-balancing mechanisms for high availability as per the design

3. Integration with Existing Infrastructure

- Integrate SD-WAN solution with existing LAN and WAN infrastructure.

4. Security

- Implement security features and functionalities mentioned in the Technical Specifications

5. Testing and Validation

- Perform testing of the SD-WAN solution
- Validate application performance, failover mechanisms, security policies, and traffic prioritization.
- Conduct user acceptance testing (UAT) with the DOT's team to ensure the SD-WAN solution meets functional requirements.

6. Training

- Provide training to the IT team on SD-WAN solution which includes product overview, design and configuration.
- Handover HLD, LLD and Standard Operating Procedures (SOPs) document.
- Conduct knowledge transfer to ensure the DOT's team can operate and maintain the solution post-deployment.

7. Support and Maintenance

- The successful bidder to deploy On-Site Technical Resource at DOT Head office for period of 1 Year to support and maintain the SD-WAN Solution along with Network Infrastructure. Job Responsibilities are mentioned in **Section 5.12.1**. DOT will discuss with Successful Bidder and decide the renewal of the Technical Engineer Services after 1 year.

3.2.1.4. Locations of the SD-WAN Devices Deployment

DOT shall share the detailed locations of all its Branch Office locations (spread across Kerala) and other required information to the successful bidder.

3.2.1.5 SD-WAN - Technical Specifications

3.2.1.5.1 Functional Requirement

Technical Specification – Functional Requirement		
Sr. No	**Requirements**	**Compliance Yes /No**
Architecture, Deployment and Performance		
1	The proposed SDWAN Solution should be implemented as true software defined WAN solution. The proposed solution should have Central Management solution/Controller/Orchestrator, Head-End and Branch Devices.	
2	The proposed SD-WAN Solution shall be implemented as secure software-defined network Architecture with a complete separation of Control, Data and Management plane.	
3	The proposed SD-WAN Solution shall integrate different types of connectivity (MPLS, ILL, FTTH, Broadband, SIM, ADSL/DSL, Ethernet, Fibre, LTE/3G/4G/5G, satellite) into a single, end-to-end framework and encrypted tunnel that uses policies to manage traffic.	
4	The communication between the central location and the branch device running on the remote entity should be secure and encrypted.	

5	The proposed SD-WAN solution must include following components in a OEM Physical Appliance Form Factor: 1. Central Management solution/Controller/Orchestrator 2. Central SD-WAN Head-End appliances 3. Branch SD-WAN appliances	
6	The proposed SD-WAN Central Management solution/Controller/Orchestrator need to be installed on-premises	
7	The proposed SDWAN solution must co-exist with existing IP/MPLS network	
8	The proposed Central Management solution/Controller/Orchestrator solution should be a single application and not combination multiple application that expected to deliver that provides global visibility and control over multiple SD-WAN appliance. It must streamline management tasks such as device configuration, policy deployment, across the network. SDWAN Management solution should provide single platform to monitor, troubleshoot and maintain the SDWAN Features.	
9	The proposed SD-WAN solution must have separation of data and management to achieve following functionality:	
	Central Management solution/Controller/Orchestrator solution should be deployed in HA at DC and DR. a. Central Management solution/Controller/Orchestrator to provide single point of entry for Configuration and Monitoring. Central Management solution/Controller/Orchestrator shall be securely accessed and shall be capable of configuration, monitoring and troubleshooting of SD-WAN devices in the branches and Datacentres. All the devices be managed from the Central Management solution/Controller/Orchestrator. Central Management solution/Controller/Orchestrator shall maintain centralized routing table, controls route advertisement as per policy, creates end to end segments on network instructs data plane to change traffic flow as per policy.	
	b. Data Plane: Forwards traffic in encrypted tunnels, applies local policy like QoS and security policies.	
	c. Control Plane: Maintains centralized routing table, controls route advertisement as per policy, creates end to end segments on network, instructs data plane to change traffic flow as per policy.	
	d. SD-WAN appliances: SD-WAN Head End Devices: The Head Devices in DC and DR will be deployed in HA mode. SD-WAN Branch Device: All the Branch SD-WAN devices will be deployed in Standalone Mode.	
10	All SD-WAN solution components including the Central Management solution/Controller/Orchestrator and Edge SDWAN Devices shall be from the same OEM	
11	The Central Management Solution/Controller/Orchestrator shall be designed to support up to 2000 devices and any required SDWAN licenses for controllers shall be provided from day-1.	
12	From the Central Management Solution/Controller/Orchestrator DOT should be able to configure Template for branch and central locations	

13	The proposed SD-WAN Solution shall support deployment of any kind of topologies such as full-mesh, hub-spoke and partial mesh as desired by DOT. It shall be possible to change network topology only by pushing policy on Central Management Solution/Controller/Orchestrator and shall not require device by device configuration. The proposed SDWAN solution should allow multiple Hub Sites (at least 5) when deployed in multiple hubs and spoke topology.	
14	The solution shall support Customized Role Based Access Control that provides only relevant information to the user based on their roles and privileges. Solution shall also provide customized role builder and support multiple administrators. Solution shall provide detailed audit trail of changes etc. The Proposed SDWAN reporting and dashboard solution should have capacity for minimum 10 concurrent login for the admin/management console or GUI without any degradation in services.	
15	The Central Management Solution/Controller/Orchestrator shall support local, external Radius Server User/Account authentication method or two factor authentications and AD integration.	
16	The proposed SDWAN solution should have the capability to do forward error correction to mitigate against packet drops automatically when SLA (packet loss, latency, and jitter) of available link(s) is/are violated for defined application/traffic.	
17	In case of failure of Central Management Solution/Controller/Orchestrator the remote device should work seamlessly and should be accessed via GUI.	
18	The proposed solution shall allow for plug and play installation of Branch devices without requiring any manual configuration at the remote location.	
19	All elements of the solution will have IPv4 and IPv6 support from Day1	
20	The SDWAN Central Management Solution/Controller/Orchestrator shall have Capability for Zero Touch Provisioning. There shall be “No impact” on Data Plane traffic in case of Controller/Orchestrator outage.	
21	The system should be able to select the optimum path based on the network parameters like Latency, Jitter, PLR and network capacity.	
22	Solution should able Granular application and service controls allow DOT to prioritize applications based on whether the application is mission-critical, latency-sensitive, or meets certain health criteria	
23	The system should be able to prioritize business critical applications and should prioritize this traffic over others during congestion	
24	The system should be able to automatically steer traffic flows to the optimum path, based on policy definition in the software defined network controller	
Dynamic Load Sharing of Traffic Based on Business and/or application policies		
25	The proposed SDWAN Head-End and Branch devices shall be able to load balance traffic across multiple connectivity paths based on load balancing algorithms efficiently using all available connectivity. System shall also detect blackouts & brownouts by supporting active-active load balancing and fast session failover. Shall be able to integrate different types of connectivity (MPLS/ILL/FTTH/Broadband/SIM/VSAT) for Active-Active	

	Load Balancing on per packet, per session and per Application basis with Seamless Failover without traffic duplication on the connected links at a site.	
26	The proposed Edge devices shall be able to identify and classify applications at OSI layer L4-L7.	
27	The proposed solution shall be able to dynamically control data packet forwarding decisions by looking at application type, performance, policies, and path status	
28	The proposed solution shall allow various traffic load balancing requirement like, between equal and unequal bandwidth links, application flow via specific link only.	
29	The SDWAN shall be able capable of Seamless Failover of Voice/Video and Data Traffic	
30	The system shall do continuous performance monitoring of latency for all network paths, links and make decisions to forward critical applications over the best performing path based on the defined application policy.	
31	The SD-WAN solution shall improve the degraded link (e.g. Forward Error Correction/SACK or equivalent, Quality of Service (QoS), Jitter buffering etc.) to maintain application performance for high priority applications	
32	In the event of link degradation or link down the solution shall be able to move traffic to a better link without any application disruption by moving the data traffic on a per packet/flow basis	
33	The proposed solution shall be able to automatically route over the overlay WAN without the need for a separate routing protocol. Solution shall support automatic overlay routing without any manual configuration.	
34	The proposed solution shall be able to prioritize business critical applications and shall have the capability of prioritizing traffic during congestion	
35	The proposed solution shall make path optimization decisions on the Edge device and not through the central controller to reduce delay and improve performance.	
36	Proposed SDWAN solution should have capability to define Application Aware Routing mandatorily in all the following manners:	
	a. Traffic for any application / group of applications should automatically select the best path, based on the real-time system calculated parameters like Latency, Jitter, Packet loss etc. The application traffic must switchover to alternate link/path even if any one of the configured parameters violate the predefined SLA, irrespective of parameter priority	
	b. There should be provision to bind an application / Group of applications to a specific link. However, the traffic bounded to a specific link should auto switchover on alternate available link in cases like Link down, Link violates any of the configured SLA, irrespective of parameter priority.	
	c. There should be option to distribute application's traffic to all the available links, for which link parameters are within defined	

37	In the proposed SDWAN solution, Head-end and Branch device should be able to monitor different path parameters for all the WAN Links in near real time.	
38	In the proposed SDWAN solution, the WAN path selection for the traffic originating from branch should be dynamically done based on the policy configured/pushed from the Central Management Solution/Controller/Orchestrator.	
39	The device should support load balancing on available WAN links based on	
	1. Session based	
	2. Packet based	
40	The solution should be able to load balance traffic among various links without traffic duplication on links and the solution shall seamlessly failover traffic on failure of primary link to secondary and tertiary links in sub second interval. The failover shall seamlessly work among various type of connectivity's e.g. MPLS / Internet / Broadband /4G LTE / 5G /VSAT etc. The seamless failure implies that sessions should not break while switching traffic from Primary to other links and vice versa.	
41	The Proposed SDWAN solution should load balance the critical application (Highest Priority) traffic through second best link and so on, if the best link bandwidth is utilized up to configured threshold.	
42	The proposed SDWAN solution should be able to load balance the traffic across multiple WAN links of uneven bandwidth. Simultaneous monitoring of the links should be done to avoid congestion on the lower-bandwidth link.	
43	The proposed SDWAN Solution should be able to utilize all types of service provider links irrespective of the last mile connectivity type (e.g., MPLS, 4G, VSAT, RF, 5G, etc.) simultaneously and load balance the traffic without traffic duplication amongst the all the links irrespective of the last mile connectivity as per configured policies. The devices should support Ethernet extender to support 4G and 5G connectivity (4G, 5G devices will be provided by Telecom Service Provider and Ethernet cable will be extended till router).	
44	The proposed SDWAN solution should continuously monitor the link flaps. If the link is not stable, then the traffic to be diverted on alternate superior link. The inferior link to be put in monitor state with a configurable time interval. Once, the inferior link is stable and within the SLA parameters configured, all existing as well as new sessions must shift to superior link accordingly as per configured policy.	
45	The proposed SDWAN solution should support seamless application accessibility across Datacentres & Branches during auto failover of WAN links and load balancing without any session disconnection and traffic duplication. The session switches over or switch back between links without any manual intervention.	
	a. Sessions should seamlessly failover to the next- superior link available without terminating the active sessions in case the performance of superior link gets degraded, without packet duplication. After restoration of the link, traffic should be moved back as per the last known good status. The seamless failover functionality shall work without traffic duplication on primary and secondary lines.	
	b. Should be able to maintain symmetric path for data traffic (send and receive	

	the packets on same link for a session) at any point of time.	
High Availability		
46	The proposed SD-WAN Central Management solution/Controller/Orchestrator must be deployed in a High Availability at DC and DR	
47	The proposed SD-WAN Head-end Devices must be deployed in a High Availability at DC and DR	
48	The SD-WAN branch office devices should be deployed in standalone mode.	
Interoperability		
49	The proposed SDWAN solution should support static encrypted tunnel using well-known global standard protocol	
	1. any designated device of same OEM	
	2. any designated device of different OEM	
50	The devices in the proposed SDWAN solution should have interoperability for IPSec, routing protocol (BGP/static/OSPF), etc. with the existing products from multiple vendors in DOT's eco-system	
51	The proposed SDWAN solution should be able to integrate with AAA, AD, SIEM and NMS systems	
52	In the proposed SDWAN solution all the devices must support Port aggregation protocol with Open standards like LACP on LAN ports.	
53	In the Proposed SDWAN Interfaces of all devices (Central Management solution/Controller/Orchestrator/Head-end Devices/Branch Devices) must support virtual/sub interface (IEEE 802.1Q) on WAN and LAN side interface.	
Security		
54	All devices provisioned as part of the solution should have the capability to restrict management access based on IP address / Subnets.	
55	All the sensitive information (including passwords, keys) should be hashed (using SHA-2 or any other higher standards) and should not be visible in plain text at CLI / GUI.	
56	The encryption shall be done as per IPSec standards using AES-256 or AES-GCM or higher coupled with SHA 256/512 Hash Algorithm	
57	The proposed SDWAN devices/appliances should have authentication and authorization only with any of the pre-configured Central Management Solution/Controller/Orchestrator which are placed in Datacentres. Data should only flow after the device authentication and authorization is done by the Central Management Solution/Controller/Orchestrator.	
58	Proposed SDWAN devices should pull/push the configuration template from/to Central Management Solution/Controller/Orchestrator after authentication and authorization only.	
59	The solution should be able to upgrade edge/branch device from authenticated & authorized Central Management Solution/Controller/Orchestrator devices only.	
60	All devices access should be available through SSH (AES-128, AES-256 and higher, SHA-2 and higher) or HTTPS (TLS 1.2, TLS 1.3 or higher) and should support all future algorithms through upgrades or updates.	
61	SD-WAN must have geo-blocking system so that traffic can be block / filter based on locations	
62	SD-WAN Devices Local console access must have password-based authentication	

63	Logs should be configured to include but not limited to following: i. Authentication failures. ii. Account created/deleted/disabled. iii. Password changes for privileged accounts. iv. Changes in configuration settings. v. Start and stop of service. vi. System/Console alerts/errors or failures/fault logs. vii. Administrator or Root user activities. viii. Access to audit trails. ix. Creation and deletion of system-level objects. x. Alarms/alerts raised by the access control system (such as violation of any ACL defined) xi. Details of system/files accessed of the SDWAN device. xii. Use of privileges. (i.e., Privilege escalation)	
64	All remote site traffic shall be encrypted when transported over WAN transport links: MPLS, ILL, FTTH, Broadband, SIM, ADSL/DSL, Ethernet, Fibre, LTE/3G/4G/5G, satellite network protecting Data Confidentiality and Integrity.	
65	The proposed Head-end and Branch office SD-WAN devices shall have security functionalities like Firewall, Web Filtering and Intrusion Prevention.	
Routing and VPN		
66	The system should ensure that any change in physical connectivity (Link 1 to Link 2 connectivity in case of multiple links being terminated on the branch device) or any change in physical connectivity type (Link 1 connectivity changed from internet broadband to MPLS or vice versa, in case of multiple WAN links being terminated on the Branch device) does not require any change in virtual private network configuration in the Central Management Solution/Controller/Orchestrator or SD-WAN Devices.	
67	The system should support encapsulation types, but not limited to following: A. IPSEC B. GRE	
68	The system should support the following minimum authentication algorithms for Data Integrity: a. SHA-512 b. SHA-384 c. SHA-256	
69	The system should support the following minimum encryption algorithms for Data Security: a. AES-128 b. AES-256	
70	The system should allow time- based/scheduled manner refresh of the encryption key / standardized encryption algorithm for each virtual private network.	
71	The system should allow for alternate hub destinations to be created for application specific traffic using a policy defined for it.	
72	The proposed SD-WAN solution shall be able to auto configure VPN tunnels without limitation of the number of remote sites. The solution shall be able to auto configure VPN tunnels of a minimum of 2000 sites with and without manual VPN tunnel configurations.	
73	In case of a change in WAN IP Address (Private/Public) the SD-WAN solution	

	shall detect and rebuild the VPN tunnel without manual VPN configurations.	
74	The proposed SD-WAN Solution shall be capable to view the routing table of the entire enterprise and have ability to audit and troubleshoot	
75	The tunnel creation should be automatic without any manual configuration on the edges and the controller.	
76	The proposed Solution should support route manipulation on LAN, WAN and Overlay to achieve various hybrid network integration requirements.	
77	The proposed Solution should support Site to Site IPSec tunnel on LAN/WAN side with any third party IPSEC capable appliance for hybrid deployments.	
78	The proposed Solution be capable of client-based VPN services	
NAT		
79	The SD-WAN solution shall support NAT and port forwarding of all LAN side prefixes.	
80	The SD-WAN solution shall support source NAT on location subnets to allow overlapping location subnet to communicate to the Datacentre. Also, the solution shall support LAN side destination NAT for simplicity of redirecting traffic	
QoS		
81	The solution shall be capable of configuring QoS weights on default applications, bespoke applications & destination IP based QoS definitions/VLAN based/VRF based etc	
82	The SD-WAN solution shall be able to apply QoS policies for all traffic types including TCP, UDP and other non-TCP traffic types	
83	The SD-WAN solution should include a QoS mechanism that is able to protect delay sensitive flows like Voice, Video.	
84	The device should support industry standard IP QoS mechanisms including (Traffic Class, IP Precedence and DSCP - Differentiated Services Code Point).	
85	QoS should work for all kind of traffic in both forward and backward direction. All the WAN ports of device should support QoS.	
86	The device should support (but not limited to) industry standard congestion management techniques such as class-based weighted fair queue, Low latency queue or equivalent.	
87	The device should support the per class bandwidth definition for QoS policy. The definition should support all the formats such as Kbps/Mbps/Gbps and percentage bandwidth. However, during configuration any one of these can be selected.	
88	The device should support separate QoS implementation (but not limited to) per wan link, per application (custom / defined), per class.	
89	The proposed SDWAN solution should be able to define minimum guaranteed bandwidth per class of service.	
WAN Optimisation		
90	Solution should have capability for WAN Optimization with following features	
	1. TCP Optimization	
	2. Sliding Window Adjustments on dynamic real time basis	
	3. UDP Optimization	
	4. Caching	
	5. Compression	

	6. Deduplication	
	7. Forward Equivalence Class	
	8. No traffic duplication for seamless failover of traffic	
91	The Solution shall have capability of Multi line WAN Optimization which means the WAN Optimization functions shall seamlessly continue on secondary line in case of primary link failure.	
92	The Solution shall have capability of Byte Level WAN Optimization for better granular performance.	
Centralized Management/Orchestrator/Controller		
93	The Centralized Management/Orchestrator/Controller shall provide a single, unified platform for network service provisioning, device configuration, monitoring and assurance and change management.	
94	The Centralized Management/Orchestrator/Controller should provide a dashboard that provides state of new appliances (Online, Offline, not connected).	
95	The Centralized Management/Orchestrator/Controller shall have web-based GUI. The solution shall support SSL for access to the management web GUI.	
96	The Centralized Management/Orchestrator/Controller should support a minimum of 2000 branch locations.	
97	The Centralized Management/Orchestrator/Controller should be able to segment the network with IP Subnets spanning across locations	
98	The Centralized Management/Orchestrator/Controller should be able to attach each local area network IP segment to a different network segment or by creating multiple virtual private networks.	
99	The Centralized Management/Orchestrator/Controller should have an option for integrated inbuilt DNS based feature for inbound load balancing of Internet and MPLS traffic and it shall be capable of Site Load Balancing in the same appliance to achieve DC/DR fail-over	
100	The Centralized Management/Orchestrator/Controller should have the capability to synchronize automatically as well as manually with geographically redundant devices. Automatic synchronization should be done in near real-time (within 60 seconds) without any manual intervention.	
101	The Centralized Management/Orchestrator/Controller should have the facility to configure, manage and upgrade the branch routers/devices centrally from orchestrator. The configuration on head-end / branch end must be minimal or equivalent to zero-touch provisioning. Configuration such as Interface IP address, orchestrator IP address to initiate initial reachability to Central Orchestrator, other parameter specific to each branch (ISP BGP neighbour IP, LAN IP etc.) can be locally done during the on boarding process of head-end/branch device. Rest all configurations to be done centrally from the orchestrator.	
102	The Centralized Management/Orchestrator/Controller should be capable of monitoring any configuration mismatch between orchestrator and branch device.	
103	The solution shall support end-to-end real-time flow visualization for the application paths for identifying issues and taking corrective actions.	

104	All network-wide configurations shall be from the centralized management/orchestrator/controller	
105	The Centralized Management/Orchestrator/Controller solution should support network wide device and network visibility for all the devices in the terminated on the devices irrespective of the type of link. Device should be equipped with the features like Visualize in real-time, Customized graphs and reports, Link Availability, WAN Link utilization, Detailed bandwidth usage of applications, Link wise WAN Link Latency, jitter and packet loss, SLA monitoring and compliance, Appliance load and utilization, Alert for high resource utilization All types of alarms, Application performance monitoring.	
106	In case of performing any trouble shoot, the solution should have trouble tracking tools such as TCP dump, ping, trace route etc.	
107	The SD-WAN Solution shall support SYSLOG export directly from the SDWAN edge device	
	The solution shall support SNMP V2/V3, SYSLOG, IPFIX or equivalent polling of the SD-WAN devices and send SNMP to equivalent traps/logs. A dedicated SDWAN/Routing MIBs shall be available to poll SD-WAN specific functionality.	
	SD-WAN Central Management platform must be able to centrally manage software updates for SD-WAN appliance, ensuring they are up to date with the latest security patches and features	
Network Monitoring, Reports, Analytics and Notification		
108	The proposed solution to provide Traffic reports per site: availability, bandwidth usage per circuit (both upstream & downstream),latency, packet loss, QoS per link etc	
109	The proposed solution to provide Application reports: Applications usages, Bandwidth Consuming Applications.	
110	The solution must support device health monitoring for all the SD-WAN Devices in the network.	
111	Status of all the SD-WAN Devices deployed in WAN infrastructure should be viewed in a Graphical Map	
112	The solution must support events and email-based alarm to notify the administrators when any device/link fault or network performance degradation happens.	
113	The thresholds should be configurable per location /per SD-WAN Device.	
114	The alerts should be received for all WAN performance related events (like change in link characteristic, bandwidth, congestion)	
115	Edge device in the proposed solution should be able to send path parameters data, session details, etc. to the central reporting device at interval of every 5-minute maximum.	
116	The Dashboard should be able to display the current load on head-end and Branch devices	
117	The dashboard of the proposed solution should be capable of doing near real-time monitoring of all the links in GUI with respect to the following (but not limited to) parameters:	
	1. Packet loss	
	2. Jitter	
	3. Link errors	

	4. Bandwidth utilization	
	5. Latency	
	6. Duplex, Speed	
	The dashboard of the proposed solution should be capable of providing QoS Reports	
	The dashboard of the proposed solution should be capable of providing Application Reports	
118	SD-WAN Central Management Platform natively should be able to deliver monitor and provide report on the basis of following: 1. Device inventory 2. Session details 3. Changes in application usage and user activity 4. Users and applications that take up most of the network bandwidth 5. Uptime and downtime 6. Network activity 7. Tunnel activity	
119	SD-WAN Central Management Platform must deliver on-box downloadable reporting capability for application performance and link performance	
120	SD-WAN Central Management Platform should natively provide reports and same should be schedule for sent via email	
121	The proposed solution must be able to generate notifications for the following: 1. Link Flaps at the remote branch location 2. ISP link quality degrades 3. Link utilization along with threshold 4. CPU, Memory and Disk Utilization of the Branch Device	
122	The software defined network controller must have REST APIs available for 3rd party integration or integration with custom automation tools.	
RMA and Support		
123	The OEM should have a Technical Assistance Centre (TAC) available 24*7 with regional support centres for quick response to incidents.	
124	The proposed solution should have Direct Support from OEM (24*7) TAC Support and NBD replacement for the fault devices.	

3.2.1.5.2 Centralized Network Orchestrator/Manager/Controller

Technical Specifications – Centralized Network Orchestrator/Manager/Controller		
Sr. No	**Requirements**	**Compliance Yes /No**
1	The proposed Centralized Network Orchestrator /Manager/Controller devices should have following minimum specifications	
	a. Minimum 4 Gbps throughput	
	b. Multicore CPU Architecture with minimum 8 Core processor.	
	c. Hardened 64-bit operating system	
	d. Minimum of 32 GB of RAM	
	e. Minimum 512 GB storage	
	f. Minimum interfaces from day 1: 4* 1G/10GCopper 4* 1G/10GFiber	

2	In case of loss on connectivity with the SD-WAN Central Management Solution/Controller/Orchestrator, the SD-WAN Head-End and branch devices should continue to use the existing policy to forward the traffic. The data plane should not be affected even for single minute, with the loss of connectivity with the controller and Data plane traffic shall work without any interruption.	
3	The proposed SD-WAN Centralized Network Orchestrator /Manager/Controller devices must be capable of handling IPv6 on inward-facing interfaces and ISP outward-facing interfaces.	
4	The proposed SD-WAN Centralized Network Orchestrator /Manager/Controller devices shall be able to support Management of minimum 500 devices from single Hardware.	
5	The proposed SD-WAN Centralized Network Orchestrator /Manager/Controller shall support high availability Architecture	
6	The Proposed SD-WAN Centralized Network Orchestrator /Manager/Controller devices shall support Geographic redundancy (500 KM) with high availability Architecture at each of the locations	
7	The proposed Centralized Network Orchestrator /Manager/Controller devices be supplied with, Power Cable and Rack Mount kit.	
8	The proposed SD-WAN Centralized Network Orchestrator /Manager/Controller should have redundant power supply with hot Swapping facility	

3.2.1.5.3 Head-End Devices

Technical Specifications – Head End Devices		
Sr. No	**Requirements**	**Compliance Yes /No**
1	The proposed Head-end devices shall be able to connect in SD-WAN Architecture and shall meet requirements as per specification mentioned under functional requirement.	
2	The proposed SD-WAN Head-end Devices in the Datacentres should be able to peer with the existing routers in the Datacentres and shall support static and dynamic routing protocols.	
3	The proposed SD-WAN DC Devices should be Rack-mountable Physical appliances and should have Control Plane separated from the Data Plane in the Device Architecture itself.	
4	The Proposed SD-WAN Head-end Devices should have following minimum specifications	
	a. Minimum 20 Gbps throughput after enabling Firewall and Intrusion Prevention functionalities.	
	b. Multicore CPU Architecture with minimum 16 Core processor.	
	c. Hardened 64-bit operating system	
	d. Minimum of 256 GB of RAM expandable to 1TB	
	e. Minimum 512 GB storage	
	f. Must support at least - 200000 concurrent sessions	
	g. Site to Site VPN: Minimum 2500	
	h. Minimum Interfaces from Day 1: 8* 1G/10 G Copper	

	4* 1G/10 G Fiber	
5	Solution should be able bundle links that allows to group multiple physical links (that different ISPs use to communicate with the same destination) into a virtual SD-WAN interface.	
6	The SDWAN Head-end device should be able to automatically load balance and fail over MPLS, Internet, Leased Lines etc without traffic duplication with seamless failover capabilities. The VoIP and Video calls call shall not disconnect on failure of primary links as the traffic shall be migrated from primary to other available links in seamless manner without traffic duplication.	
7	In case of loss on connectivity with the Central Management Solution/Controller/Orchestrator, the proposed SD-WAN Head-End device should continue to use the existing policy to forward the traffic. The data plane should not be affected with the loss of connectivity with the controller and Data plane traffic shall work without any interruption.	
8	The proposed Head-end devices must be capable of handling IPv6 on inward-facing interfaces and ISP outward-facing interfaces. Dual stack support required from Day1.	
9	The proposed Head-end devices shall support failover of less than 20 Milliseconds.	
10	The proposed Head-end devices should be supplied with Power Cable and Rack Mount kit.	
11	The proposed Head-end devices should have integrated inbuilt DNS based feature for inbound load balancing of Internet and MPLS traffic and it shall be capable of Site Load Balancing in the same appliance to achieve DC & DR fail-over	
12	The proposed Head-end devices should have capability for WAN Optimization with following features	
	1. TCP Optimization	
	2. Sliding Window Adjustments on dynamic real time basis	
	3. UDP Optimization	
	4. Caching	
	5. Compression	
	6. Deduplication	
	7. Forward Equivalence Class	
	8. No traffic duplication for seamless failover of traffic	
	9. Forward Error correction	
13	The proposed Head-end devices shall have capability of Multi line WAN Optimization which means the WAN Optimization functions shall seamlessly continue on secondary line in case of primary link failure.	
14	The proposed Head-end devices shall have capability of Byte Level WAN Optimization for better granular performance.	
15	The proposed Head-end devices should support NAT IPV4 to 6 and vice versa for 200000 concurrent sessions.	
16	The proposed Head-end devices should have Redundant Power Supply Units Hot Swappable.	

3.2.1.5.4 Branch Devices - Type 1: District Treasury Offices

Technical Specifications Branch Devices – Type 1: District Treasury Offices		
Sr. No	**Requirements**	**Compliance Yes /No**
1.	The proposed Branch devices shall be able to connect in SD-WAN Architecture and shall meet requirements as per specification mentioned under functional requirement.	
2.	The proposed SD-WAN Branch Devices should be Rack-mountable Physical appliances.	
	a. Minimum 200 Mbps throughput after enabling Firewall and Intrusion Prevention functionalities.	
	b. Multicore CPU Architecture with minimum 8 Core processor.	
	c. Hardened 64-bit operating system	
	d. Minimum of 32 GB of RAM and the appliance	
	e. Minimum 64 GB storage	
	f. Minimum Interfaces which are as follows 2* 1G/10 G Copper 4* 1G/10 G Fiber	
	g. Must support at least - 100000 concurrent sessions	
	h. Site to Site VPN: Minimum 100	
3.	The proposed SD-WAN Branch Devices should be able to bundle links that allows to group multiple physical links (that different ISPs use to communicate with the same destination) into a virtual SD-WAN interface	
4.	The SDWAN Branch devices should be able to automatically load balance and fail over VoIP, VPN, and Endpoints across divergent media types of MPLS, Internet, Leased Lines etc without traffic duplication with seamless failover capabilities i.e. the VoIP and Video calls call shall not disconnect on failure of primary links as the traffic shall be migrated from primary to other available links in seamless manner without traffic duplication.	
5.	It should be possible to configure the LAN side subnets at branch location SD-WAN devices centrally from the Central Management Solution/ Controller/ Orchestrator.	
6.	In case of loss on connectivity with the Central Management Solution/Controller/Orchestrator, the SD-WAN Branch Office devices should continue to use the existing policy to forward the traffic. The data plane should not be affected with the loss of connectivity with the controller and Data plane traffic shall work without any interruption.	
7.	The proposed SD-WAN Branch Devices must be capable of handling IPv6 on inward-facing interfaces and ISP outward-facing interfaces. Dual stack support required from day 1.	
8.	The proposed SD-WAN Branch Devices shall support less than 20 Milliseconds failover of connected last mile.	
9.	The proposed SD-WAN Branch Devices should be supplied with Power Cable and Rack Mount kit.	

10.	The proposed SD-WAN Branch Devices should have minimum 1 USB Dongle	

3.2.1.5.5 Branch Devices - Type 2: Sub-Treasury Offices

Technical Specifications Branch Devices – Type 2: Sub Treasury Offices		
Sr. No	**Requirements**	**Compliance Yes /No**
1.	The proposed Branch devices shall be able to connect in SD-WAN Architecture and shall meet requirements as per specification mentioned under functional requirement.	
2.	The proposed SD-WAN Branch Devices should be Rack-mountable Physical appliances.	
	a. Minimum 100 Mbps throughput after enabling Firewall and Intrusion Prevention functionalities.	
	b. Multicore CPU Architecture with minimum 8 Core processor.	
	c. Hardened 64-bit operating system	
	d. Minimum of 16 GB of RAM and the appliance	
	e. Minimum 64GB storage	
	f. Minimum Interfaces which are as follows 6* 1G/10 G Copper	
	g. Must support at least - 50000 concurrent sessions	
	h. Site to Site VPN: Minimum 100	
3.	The proposed SD-WAN Branch Devices should be able to bundle links that allows to group multiple physical links (that different ISPs use to communicate with the same destination) into a virtual SD-WAN interface	
4.	The SDWAN Branch devices should be able to automatically load balance and fail over VoIP, VPN, and Endpoints across divergent media types of MPLS, Internet, Leased Lines etc without traffic duplication with seamless failover capabilities i.e. the VoIP and Video calls call shall not disconnect on failure of primary links as the traffic shall be migrated from primary to other available links in seamless manner without traffic duplication.	
5.	It should be possible to configure the LAN side subnets at branch location SD-WAN devices centrally from the Central Management Solution/ Controller/ Orchestrator.	
6.	In case of loss on connectivity with the Central Management Solution/Controller/Orchestrator, the SD-WAN Branch Office devices should continue to use the existing policy to forward the traffic. The data plane should not be affected with the loss of connectivity with the controller and Data plane traffic shall work without any interruption.	
7.	The proposed SD-WAN Branch Devices must be capable of handling IPv6 on inward-facing interfaces and ISP outward-facing interfaces. Dual stack support required from day 1.	
8.	The proposed SD-WAN Branch Devices shall support less than 20 Milliseconds failover of connected last mile.	

9.	The proposed SD-WAN Branch Devices should be supplied with Power Cable and Rack Mount kit.	
10.	The proposed SD-WAN Branch Devices should have minimum 1 USB Dongle	

3.2.2. Network Monitoring cum Helpdesk Platform

DOT is seeking to implement a robust Network Monitoring cum Helpdesk Platform which will serve as a centralized Network Monitoring to unify operations and Helpdesk Platform to track, resolve End User queries.

Note: In line with the requirement, The DOT has envisaged Network Monitoring cum Helpdesk Platform Components, Deployment Approach and Project Scope which are defined in detail under **Section 3.2.7**. The Network Monitoring cum Helpdesk Platform Components, Deployment Approach and Project Scope are not limited to the details provided below and may change from OEM to OEM. DOT reserves the right to ask Bidder along with OEM to conduct POC and demonstrate all or some of the points mentioned in the technical specification section.

3.2.2.1 Network Monitoring cum Helpdesk Platform Components

- **Network Monitoring cum Helpdesk Platform installed at DC shall have the following components**
  a. Network Fault Monitoring & Performance Management with Reporting
  b. Network Traffic Analysis
  c. Helpdesk System, Incident, Problem, and Knowledge Management
  d. Log Management
- **Additional Components**
  a. Professional Display Monitor
  75” Professional Monitor for displaying configured Monitoring cum Helpdesk Platform Dashboards in HQ
  b. Desktop
  Desktop with latest High-End configuration for the On-Site Technical Engineer

NOTE: The above are the components which are envisaged by DOT, the Bidder & OEM must consider & factor all necessary licenses to accomplish the above requirements of this RFP completely.

3.2.2.2Details for Network Monitoring cum Helpdesk Platform Deployment

DOT shall share the details of Network Infrastructure for Integration to the successful Bidder.

3.2.2.3Deployment Approach and Project Scope for Network Monitoring cum Helpdesk Platform

The selected Bidder and OEM will collaborate with DOT team to design Network Monitoring cum Helpdesk Platform that aligns with DOT’s business objectives and technical requirements. This includes but not limited to the following.

1. Assessment and Design

1.1 Requirement Analysis

- Gather and document business, operational, and technical requirements.
- Assess current Network infrastructure for Integration.

1.2 Solution Design

- Develop a detailed design document outlining system Architecture and integrations.

- Ensure scalability, security, and compliance are integral to the design.
- Ensure the proposed Network Monitoring cum Helpdesk Platform meets the requirements as mentioned in the Technical Specifications **Annexure 7(C**) while designing the solution for implementation.

2. Installation

2.1 Software Deployment

- Supply, install, and configure Network Monitoring cum Helpdesk Platform at DC in a Standalone Mode.

2.2 Configuration

- Set up Network Monitoring cum Helpdesk Platform modules, dashboards, and administrative controls. Configure and ensure network devices alert the centralized Monitoring platform events, so as to reflect real status of all network infrastructures across DOT.
- Configure and ensure that the Network Monitoring cum Helpdesk Platform should poll Network Devices at regular intervals to determine their status and working.
  Configure Helpdesk Desk tool as per the requirement

3. Integration

3.1 System Integration

- Integrate Network Monitoring cum Helpdesk Platform with Network Infrastructure for monitoring

4. Testing and Validation

4.1 Functional Testing

- Test Network Monitoring cum Helpdesk Platform functionalities against defined requirements.

5. Training and Documentation

5.1 Training Sessions

- Provide training to the IT team on Network Monitoring cum Helpdesk Platform which includes product overview, design and configuration.
  Handover HLD, LLD and Standard Operating Procedures (SOPs) document.
- Conduct knowledge transfer to ensure the DOT's team can operate and maintain the solution post-deployment.

6. Support and Maintenance

- The successful bidder to deploy On-Site Technical Resource at DOT Head office for period of 1 Year to support and maintain the Network Monitoring cum Helpdesk Platform along with Network Infrastructure. Job Responsibilities are mentioned in **Section 5.12.2**. DOT will discuss with Successful Bidder and decide the renewal of the Technical Engineer Services after 1 year.

3.2.2.4. Network Monitoring cum Helpdesk Platform - Technical Specifications

Technical Specifications - Network Monitoring and Helpdesk Platform		
Sr. No	**Requirements**	**Compliance Yes /No**
A	The proposed Infrastructure Management System Platform should be from single OEM and shall support following functionalities:	
	1. General Requirements	
	2. Network Fault Monitoring & Performance Management with Reporting	
	3. Network Traffic Analysis	
	4. Helpdesk System, Incident, Problem, and Knowledge Management	
	5. Log Management	
1	**General Requirements**	

	Refer "It" as "The Proposed Network Monitoring cum Helpdesk Platform " in the below mentioned points	
1.1	Supply, Install, Integrate, Operate & Maintain the offered Network Monitoring cum Helpdesk Platform for 5 years. It should provide a unified reporting dashboard, integrated topology and event view.	
1.2	It should have a secured single sign-on and unified console for all functions of components offered for seamless cross functional navigation & launch for single pane of glass visibility across multiple areas of monitoring & management.	
1.3	It should be accessible via a Web based GUI console/portal from intranet as well as from internet.	
1.4	It should not come with any hidden license restriction to access services/features of proposed solution i.e. all the mentioned services of solution should be available from Single OEM.	
	There should not be any restriction on user creation and concurrent user sessions in the solution for any module except mentioned explicitly.	
1.5	It should have dual-stack IP support (support both IPv4 and IPv6) and should be completely vendor-agnostic in nature to be able to monitor a multi-vendor environment	
1.6	It should facilitate at least 3 concurrent user sessions for moderation of tickets/service requests.	
1.7	It should have Push API's using Web Socket to push data to 3rd Party Application for Dashboard and Reports (to optimize bandwidth usage).	
1.8	It should have Node Tags for device grouping and resource/interface tagging for element grouping. Apart from Node Tags additionally system should have options to do device grouping based on default fields and customer fields	
1.9	It should have reporting function to allow for building various reports and track the performance of Infrastructure and operational views in real time.	
1.10	It should have web-based Reporting Tool/Layer across various functional modules (such as fault, performance, etc.) proposed Platform. The Reporting Tool/Layer should provide ability to create, edit, and distribute report templates as well as on demand reporting to quickly design, save & share ad-hoc queries. It should create topology maps containing network devices discovered while generating the reports.	
1.11	It should support creating views to quickly extract the necessary info about events based on different predefined filters or search conditions.	
1.12	It should support Keyboard shortcuts to navigate through different modules and perform faster actions	
1.13	It should provide option for the users to choose personalized GUI Colour theme (Multi-Colour, Dark/Light Theme),	
	Multiple Logo supports based on the user state, system should load their state default logo.	
1.14	It should have an IT Asset Management (IT and NON IT) and IP-Address Management features. In future without having to do any additional installations the ITAM and IPAM features can be used by buying and enabling the license.	

1.15	It should have a bi-directional integrated Network Configuration and Change Management (NCCM) & Network Detection & Response (NDR) tool with option to use NDR & NCCM features in future easily by buying and enabling the license for it without having to do any additional installations. The integration should allow assets and topology to sync from the NMS module to the NDR & NCCM features for helping in Root-Cause-Analysis of faults.	
2	**Refer "It " as "Network Fault Monitoring & Performance Management with Reporting" in the below mentioned points**	
2.1	It must monitor the performance across heterogeneous networks. It should automatically discover different type of heterogeneous devices (all SNMP supported devices i.e. Router, Switches, SD-WAN) and map the connectivity between them with granular visibility up to individual ports level. The tool shall be able to assign different icons/ symbols to different type of discovered elements. It should show live interface connections between discovered network devices.	
2.2	It must monitor and manage the devices with following attributes	
	a. Availability	
	b. Uptime	
	c. Network Latency	
	d. Usage/Throughput, Bandwidth	
	e. Error and Failure which include packet losses, interface failures, network failures etc.	
2.3	It should have a graphical topological display of all discovered network devices in real time.	
	The new devices must get added as an incremental discovery/partial without causing additional overhead of scanning the entire network. Further, any changes to devices must be identified in near real time during discovery scans	
2.4	It should support various discovery protocols to perform automatic discovery of all L2, L3 Network devices across DOT and any further Network connectivity's planned in future	
	System should have option for multiple options for discovery including IP address-based discovery, IP address range discovery, CSV based discovery for bulk discovery and it should allow options to add custom fields to support customer specific data to upload during discovery	
2.5	It shall be able to discover Ipv4 only, Ipv6 only as well as devices in dual stack. In case of dual stack devices, the system shall be able to discover and show both Ipv4 and Ipv6 IP addresses.	
2.6	The discovery must also support device redundancy discovery in case of virtual IP addresses using vendor specific protocols such as VRRP and HSRP.	
2.7	Discovery has to work intelligently by identifying the device in the network by the given IP range and categorize into network devices and servers with vendor and model details.	
2.8	Automatically learn devices that supports SNMP, HTTP, Ping, SMTP, POP3, WMI, JMX, SOAP, REST API, PDC, SSH and Telnet along with any required protocol to communicate to the devices.	

2.9	It must provide a detailed asset report, organized by vendor name, device type, listing all ports for all devices. The proposed Network Fault Management console must also provide asset inventory reports	
2.10	It must provide reports to identify unused/dormant/administratively down Network ports in order to facilitate capacity planning	
2.11	It should be an integrated solution comprising of Fault, Performance Management features from including device fault and performance, Interface Faults and Performance, Bandwidth Analysis, Response time Analysis. It should proactively analyse problems to improve network performance.	
2.12	It should have extensive reporting facility, providing the ability to format and present data in a graphical and tabular display.	
2.13	It should be able to automatically generate a notification in the event of a link failure to ensure proper handling of link related issues. It should have a provision for suppression of maintenance alarms during the maintenance period.	
2.14	SLA calculation / Isolation report should be made with the consideration of both the Primary and Secondary link together instead of individual link based. The downtime calculation will be measured when both the links are down for internal reporting and link based for ISP reporting. System should provide the flexible configuration in UI itself based on user needs	
2.15	It should provide a notification mechanism that allows administrator to define what notification channel to be used in different time of days, and able to trigger multiple notifications to alert multiple person and actions	
2.16	It should allow QoS monitoring of WAN links across multiple technologies like Cisco IPSLA, Juniper RPM, Huawei NQA etc. across multiple protocols like HTTP, TCP, FTP, DNS etc.	
2.17	QoS parameters should include link response time, link-level latency, link-level packet loss, link-level jitter, Round-Trip-Time etc.	
2.18	It must support out of the box trend reports on group of metrics or group of devices in a single report. This will help understand the performance of multiple devices against a KPI. It must support of the box capacity planning reports that assist in the analysis of capacity needs based on projected load.	
2.19	It should be MIB-2 compliant by default. Moreover, NMS, should have ability to load the vendor specific MIBs. Furthermore, NMS should have capability to use multiple MIBs at a time for getting desired information, without the use of Software coding. Data Collection should be possible on MIB2 Expressions using any OID (Object Identifier)	
2.20	It shall be possible to use a KPI at different network element levels (individual Network Device, Interface, Group of Network Devices, Links, etc.) and time dimensions out of the box modifying the KPI definition. The system must support complex KPIs that are constructed by weighted combination of other KPIs.	
2.21	It should be capable to automatically send the reports through e-mail to pre-defined user with pre-defined interval.	
2.22	It should have capability to exclude the planned-downtimes and others for SLA purpose or necessary customization should be done to achieve the requirement.	

2.23	The performance management system shall be able to collect and report data like: Packet delay and packet loss, User bandwidth usage rate, Network availability rate, CPU usage rate, Input/output traffic through physical ports, Input/output traffic through logical ports	
2.24	It should support global threshold, and it should have option to define individual resource/interface statistics level threshold	
2.25	It should have threshold & Alerts: define and store static & dynamic resource utilisation thresholds, define and maintain availability thresholds for individual components, services or applications, individual service actions / automated operations (performing a transaction, running an automated procedure etc.), support comparison of actual availability, actual performance with agreed levels and alerting on breaches	
2.26	I It should have configurable parameters like frequency, data duration, resolution duration, reset points.	
2.27	All thresholds should have set point, reset point, polarity, set point message and reset point message for ease of use.	
2.28	It should support separate Rule Engine based alarms apart from the generic threshold.	
	a. Should have capability to configure Device Group based, Node Based, Resources/Interface based, and Aggregation link based.	
	b. On Selection of Nodes/Resources/Aggregation links it have flexibility to filter based on fields available in node information	
	c. Rules should have option to apply configuration on top of performance value or based on configured threshold alarms	
	d. Rules should have option configure the breach based on min, max and average values	
	e. Should have option to configure rules and repeat counters	
	f. Should have options to select custom alarm and clear alarm messages for individual configured rules	
	g. Should have option to send severity levels like error, warning and information	
	h. Notifications support based on configured rules	
2.29	It supports sending alert via E-mail, SMS, Execute Batch file, SNMP Trap, XML notification, Pop-up window and Audio alert	
2.30	It should have Capacity Reservations: It should allow management of resource allocations and reservations (for services, applications or other needs), identify resource shortages and provide information for further analysis or procurement	
2.31	It should have Event Record & Classification: possible to generate event for all the monitoring devices, tool be used to define thresholds to generate events, collect from 3rd party using REST API , on regular interval Polling API and collect events from 3rd party system, classify them , assign different levels of severity to events, attribute events to specialised practices such as performance and capacity management, availability management, information security management, allow the automatic recording of event messages reported by external tools, filter & differentiate between different types of events	

2.32	It shall monitor all traffic from all the interfaces of the network device. Provides traffic Utilization based on individual interface level, nodes level or based on the group by location, branch, and departments as an Avg, Min and Max bandwidth, utilization, throughput or any custom monitoring parameters.	
2.33	It should have capability to configure business, non-business hours or custom time polling. These configurations should be available for every device as well as every component in the device.	
2.34	It should have the provision to disable and enable the polling of specific type of devices	
2.35	It should have capability to configure the maintenance period for any device. When device is in maintenance period there is no polling done and the SLA clock on the device is stopped.	
2.36	It should automatically generate daily reports that provide a summary of the Network Infrastructure as well as custom Reports and that are automatically sent by email at a pre-defined schedule to any recipient or save into any specific folder or drive.	
2.37	It should support instant diagnosis of the node status through Ping, Telnet and SNMP walk	
2.38	It should support Real-Time report generation for checking continuous reachability of target device	
2.39	It should provide many different types of topology representation, To perform the following:	
	1. Display physical connections of the different devices being monitored in the system	
	2. Display flat maps of the entire network or networks in a single view	
	3. Display customer maps based on user configurations	
	4. Display maps based on geo locations	
2.40	It should automatically learn IP Networks and their segments, LANs, hosts, switches, SD-WAN, etc. and to establish the connections and to correlate	
2.41	It should provide provision to draw & map user specific network diagram. It should have Integrated Web based feature to build Network Diagram	
2.42	It should be a Drag & Drop based Network Diagram builder, Dynamically Upload Images, Customizable objects to support multiple vendors, capability to export maps in an XML format and upload to any other system.	
2.43	It should provide Panel View of devices	
	a. Panel view should look similar to the actual device front panel	
	b. System should automatically detect the device model display the right panel without any additional configuration	
	c. Panel should show all the monitored interface with status	
	d. Fan status with live fan icon and LED status for power	
2.44	System should support VM, Hypervisor and Cluster monitoring from different vendors like VM-Ware, Citrix, Nutanix, Linux etc.	
2.45	System licensing should be based only on Physical Hosts and not charge separately for individual guest VMs running on VM Hosts	
3	**Refer "It " as “Network Traffic Analysis" in the below mentioned points**	

3.1	It should be able to monitor network traffic by capturing flow data from network devices. It shall be able to capture, track & analyse traffic flowing over the network via different industry standard traffic capturing methodologies viz. NetFlow, jflow, sFlow, IPFIX etc.	
3.2	It should identify which users, applications, protocols, countries, AS numbers, top routers, and top interfaces are consuming the most bandwidth	
3.3	It should have capability to alternatively capture traffic data via packet capture.	
3.4	It should be able to receive flows from non-SNMP-enabled devices, like VMware vSwitch	
3.5	It should monitor Type of Service (ToS), Differentiated Services Code Point (DSCP), and Per-Hop Behaviour (PHB), BGP AS and NEXT HOP	
4	**Refer "It " as "Dashboard" in the below mentioned points**	
4.1	It should provide Flexible, role-based dashboards and operational consoles for a common understanding of device status, risks and quality problems.	
	This should give the right experts actionable information so IT executives can focus operations priorities, project owners understand IT status, and operations staff always knows which components in the network domains are impacted and can proactively mitigate issue.	
5	**IT Service Management (Helpdesk System, Incident, Problem, and Knowledge Management)**	
5.1	Helpdesk should support following operations:	
	Log incidents /Auto logging /issues as service requests and provide a unique service request number.	
	Acknowledgement would be sent to the user along with the service ticket number through an email immediately on issue logging.	
	All issues logged would be assigned a severity level (L1/L2 or L3). Indicative severity level definitions shall be discussed and finalized in consultation with the DOT at the time of deployment.	
	The Helpdesk application would provide workflow and hierarchy through which each incident should move based on Incident severity, classification and owner.	
	The Helpdesk staff shall have provisions through the application for coordinating with concerned vendor in case issues are pertaining to any external entity product/support like:	
	a. OEM team	
	b. Network bandwidth Provider	
	c. End User Devices support provider	
	Trouble Ticketing	
	a. User Interface to log tickets and complete the process	
	b. Workflow for the Incident management process with escalation mechanism (escalation hierarchy) and approval processes	
	c. Defining SLA as per agreed solution for resolving tickets and supporting auto-escalation of the tickets based on priority and SLA	
	Reporting	
	a. Provide Real-time Dashboard for reports	
5.2	Helpdesk Tool should provide REST APIs to integrate with IT Infrastructure Management, Configuration Management, Network Management, CRM, SIEM tools to automate Events to Ticket.	

	The proposed tool must provide Dynamic REST/WEB API integration without writing any code, just by configuration on GUI with various inbuilt Authentication method.	
	Should be able to do 2 Way Integration with 3rd party and should be able to exchange the data at any stage of the lifecycle of the Event, Incident, Problem, Change, Request, Task, Release etc.	
5.3	Helpdesk Tool must provide intelligent Email-to-Incident feature. Tool should merge all subsequent email communication for a particular email-to-incident ticket into the same ticket in the form of a message thread. Tool should be intelligent enough to understand email conversation chains for merging emails to a particular incident.	
5.4	Tool should be able to provide real-time Email, SMS Notification alerts to notify respective users about any changes in ticket state and status. Tool should provide Email Communication Interface to allow technicians to send replies to customers / end users from the tool GUI and Record all the Email Communication in Chronological Order	
5.5	The tool should provide a self-service portal where End-users / requesters can log in and raise incidents	
5.6	The proposed tool must provide GUI interface for users, requesters, customers, support staff, 3rd party vendors, Field Engineers, Site Engineers, Supervisors, Managers, etc. with options to restrict amount of information that can be accessed by each role.	
5.7	The integrated ITSM module should have its own Android & IOS app	
5.8	Tool should have option to define the workflow for each service created in the Service Catalog for each process (Incident, Problem, Request)	
5.9	End user should also be able to browse the service catalogue and should be able to raise request by providing the input to particular service request, Address etc.	
5.10	The proposed tool must provide an option for users to track their incidents status, check knowledge articles and chat with technicians on particular incidents	
5.11	Users must have an option to Log, Classify, Categorize & Prioritize Requests	
5.12	The proposed tool must record Problem Record Date and Time, Problem Source, Contact Detail, Symptoms and Status	
5.13	Tool should allow Problem records to be classified according to priority and category	
5.14	Tool should allow a problem record to be escalated based on pre-established rules with option to manually overridden conditions	
5.15	The proposed tool must have a powerful knowledge management functionality	
5.16	In FAQ/Solutions type of knowledge, system should allow to add multiple questions/multiple solutions with single knowledge article	
6	**Log Management**	
6.1	The proposed solution should have option to collect and store system logs from target devices including firewalls, routers, switches, WLC, servers, SD-WAN devices, applications & databases	

6.2	The proposed solution should have multiple filtering options for incoming system logs based on target device, log_ID, severity, level, message, OS type, application / database etc	
6.3	The proposed solution should have an option to export specific syslog messages to users via email / SMS	

4. Instructions to Bidders and OEM's

4.1. Project Schedule and Payment Terms

4.1.1 Project Schedule

The Project Schedule is a critical element of this tender, serving as a roadmap for the successful implementation of the proposed solutions. The following project schedule outlines the key milestones, deliverables, and deadlines, ensuring that the implementation progresses in a structured manner. Adherence to the mentioned timeline is essential to prevent delays.

SD-WAN

Sl. No.	Activities	Details
1	T0 → Delivery of Items	T0 = 8 Weeks from the Date of PO
2	T1→ Completion of SD-WAN Appliance Mounting, Connectivity and Power on Self-Test (POST) Completion of Network Monitoring cum Helpdesk Platform Deployment, Configuration	T1 = T0+20 Weeks
3	T2→ Completion of SD-WAN Solution and Network Monitoring cum Helpdesk Platform Advanced Configuration, Integration and Testing	T2 = T1+8 Weeks
4	T3→ SD-WAN and Network Monitoring cum Helpdesk Platform User Acceptance Test	T3 = T2+6 Weeks
5	T4→ SD-WAN and Network Monitoring cum Helpdesk Platform Training	T4= T3+2 Weeks
6	T5→ Project Sign-Off and Handover to Operation Team	T5 = T4+2 Weeks

4.1.2 Payment Terms

The DOT will make the payment subject to signing of the contract as follows:

Sl. No.	Milestone	Percentage
1	Supply, Initial Installation of the SD-WAN Appliances and Network Monitoring cum Helpdesk Platform	50%
2	Completion of SD-WAN AND Network Monitoring cum Helpdesk Platform	40%

	Deployment and UAT	
3	This remaining payment will be released on quarterly basis in equal instalments over a period of 5 Years (20 Quarters), after deducting the Penalty for previous quarter, if any.	10%

- The DOT will release payment within 15 days from the date of receipt of invoice. In case of Dispute/s, payment will be made within 15 working days of resolution of dispute/s. No penal Interest will be paid for delayed payment.
- Payment will be released by the DOT after deduction of applicable taxes at source of the agreed payment to the bidder (for which contract will be executed) in stages on completion of the activities as per the RFP. All out of pocket expenses, travelling, boarding and lodging expenses for the entire term of this RFP and subsequent agreement is included in the amounts and the bidder shall not be entitled to charge any additional costs on account of any items or services or by way of any out-of-pocket expenses, including travel, boarding and lodging etc.
- The original invoice submitted shall be in the name of the DOT and the name of the consignee shall also be mentioned in it.

4.2. Performance Security

- The performance security amount is 3% of the Contract Awarded Value (For Successful H1 Bidder).
- There will be a performance security deposit (Performance Bank Guarantee) amounting to the total value excluding taxes, which shall be submitted by the successful bidder to the DOT within 14 working days from the date of issuance of 'Letter of Intent' from DOT.
- The contract duly signed and returned to the DOT shall be accompanied by a demand Draft or Bank Guarantee.
- Upon receipt of such contract and the performance security, the DOT shall issue the Supply Orders containing the terms and conditions for the execution of the order.
- Failure of the successful bidder in providing performance security and/or in returning contract copy duly signed in time shall make the bidder liable for forfeiture of its EMD.

4.3. Signing Authority

Bidders should ensure that the tender documents/offer/ tender acceptance letter and all other related documents has been signed by appropriate/authorized representative of the Bidder. Withdrawal of offer/non-acceptance of orders placed based on the offers submitted by bidders on their letter head will not be allowed on the ground that offer was not signed by authorized person.

4.4. Bid Validity

The proposal submitted shall indicate that it is a firm and irrevocable offer and shall remain valid for a period of not less than 180 Days from the date of bid opening. All the responses including the quotes would be binding on the Bidder for the specified time period.

4.5. Deadline for Submission of Bids

- The purchaser may, at its discretion, extend this deadline for the submission of bids by amending the bidding documents, in which case all rights and obligations of the purchaser and bidders which were subject to the previous deadline will thereafter be subject to the deadline as extended.

- It is the responsibility of the bidders to ensure that the completed bidding documents are delivered to the DOT before the closing date and time stipulated above for receipt of bid, failing which the bid would be considered late and rejected.

4.6. Bid Submission

Bidders must submit their bids along with supporting documents, to support their eligibility, as required in this tender document through www.gem.gov.in. No manual submission of bid is allowed, and manual bids will not be accepted under any circumstances.

4.7. Right to accept any bid and to reject any or all bids

Notwithstanding anything to the contrary contained herein, DOT reserves the right to accept or reject any Bid and to annul the bidding process and reject all bids at any time prior to award of Purchase Order.

4.8. Opening of Technical Bid

The technical bids will be opened, evaluated and short listed as per the Eligibility and Technical Qualification. All documents in support of Eligibility and Technical Qualification shall be submitted online. Bids short-listed by this process will be taken up for opening the commercial bid.

4.9. Technical Presentation

The DOT will request the Technically Qualified OEMs for the On-line presentation of Proposed SD-WAN Solution and Network Monitoring cum Helpdesk Platform. The Agenda for Technical presentation to include technical capabilities, proposed components and deployment approach in line with RFP requirement for SD-WAN Solution and Network Monitoring cum Helpdesk Platform.

4.10. Opening of Commercial Bids

Technically Qualified Bidders who score 50 Marks or more shall only be considered for opening and evaluation of the Commercial Bid.

4.11. Award of Contract

The contract will be awarded to the Highest scoring bidder as per the QCBS evaluation criteria after scrutiny of the technical bids, presentation and price bid opening. However, the DOT reserves the right to reject the claims of the lowest scoring evaluated bidders.

4.12. Notification of Award/Letter of (LOI)

- The DOT will notify the successful bidder(s) by email indicating there in the essential details like proposed solution and corresponding prices accepted. This notification is undertaken by issuing a Letter of Intent (LOI) by the DOT.
- The successful bidder, upon receipt of the LOI, shall furnish the required performance security and submit an agreement in the prescribed format within 14 working days, failing which the EMD will forfeit, and the award will be cancelled.
- The Notification of Award shall constitute the conclusion of the Contract.

4.13. Responsibility of Successful bidder

The following works are the responsibility of the successful bidder and OEM but not limited to the following.

- Once the contract is awarded, the successful Bidder must submit the Purchase order acceptance copy. Bidder and OEM to share design document within 4weeks after getting the Purchase Order / Work Order.
- Bidder and OEM to coordinate with other dependency/related services as required to gather the required information.
- Installation and Configuration of the proposed SD-WAN solutions and Network Monitoring cum Helpdesk Platform.
- Provide Training to DOT.
- Handing over of the Solutions to the DOT representative / operational team.
- The bidder to Deploy Technical Engineers for SD-WAN Solution &Network Monitoring cum Helpdesk Platform respectively for a period of 1 year for management and operations.

5. Terms & Conditions

5.1. Tender Document Clarification & Amendment

Amendment of Tender Document

At any time prior to the last date and time of receipt of bid, DOT may, for any reason, whether at its own initiative or in response to a clarification requested by prospective Bidder(s) may modify the Tender Document by an amendment. In order to accord prospective Bidder(s) reasonable time to prepare their bid, DOT may, at its discretion, extend the last date and time for submission of Bid.

Completeness of Response

Bidder(s) are advised to study all instructions, forms, terms, requirements and other information in the RFP documents carefully. Submission of bid shall be deemed to have been done after careful study and examination of this RFP document with full understanding of its implications.

Response to this RFP should be full and complete in all respects. Failure to furnish all information required or submission of a proposal not substantially responsive in every respect will be at the Bidder's own risk and may result in rejection of their bid.

Proposal Cost

DOT shall not be liable for any cost incurred by bidder(s) in preparing responses to this RFP or for any work performed prior to official appointment by DOT.

Language

The Bidder(s) shall quote the rates in English language. The rate shall be in whole numbers.

Rectification of Errors

Arithmetical errors in the Commercial Bid will be rectified on the following basis.

- If there is a discrepancy between the unit price and the total price that is obtained by multiplying the unit price and quantity, the unit price shall prevail and will be considered for future calculations.

Note: If Bidder does not accept the correction of errors, their bid will be rejected.

5.2. Bids Rejection Criteria

At any stage of this tender evaluation, the DOT shall be fully entitled to disqualify any bidder or for any reason whatsoever including but not limited to the following general reasons.

General Rejection Criteria:

- Failure to submit the requisite information (including any additional information requested by DOT) and supporting documents within the required timeframe.
- Bids submitted with insufficient documentary proofs also will be rejected.
- Bids received through any other form other than through GEM portal – gem.gov.in
- Documentary inconsistencies in the information submitted.

- Wilful misrepresentation in any document provided.
- Bids without signature of person (s) duly authorized on required pages of the bid and all pages of Tender Documents
- Bids without Authorisation letter for signing authority (or) power of attorney or its certified true copy.
- Any effort on the part of a Bidder to influence the bid evaluation, bid comparison or evaluation results or decisions including the false communication w.r.t the proposed solutions.
- If the information provided by the Bidder is found to be incorrect / misleading / fraudulent at any stage / time during the evaluation process
- Hand-written Documents, Erasures or Alterations. The offers containing erasures or alterations shall not be considered. All hand-written material, corrections or alterations as part of the submittals shall be rejected outright.
- Bids without signature of person (s) with stamp duly authorized on required pages of all the bid and tender document pages.

Note: In the event DOT disqualifies any bidder or applicant on above conditions, it shall not be required to disclose the reason for such disqualification.

Technical Rejection Criteria

- Bids submitted with insufficient documentary proofs.
- Technical bid should clearly specify proposed Hardware and Software for the proposed solution.
- Information provided by the Bidder is found to be incorrect / misleading to DOT / Management at any stage or time during the tendering process.
- Technical Bid containing commercial details or its indications.
- Revelation of prices in any form or by any reason before opening the commercial bid.
- Failure to furnish all information required including the eligibility criteria and similar project experience by the tender document or submission of a bid not substantially responsive to the tender document in every respect.
- Failure to provide adequate technical proofs for the compliance statements.
- Bidders not complying with the technical and general terms and conditions as stated in the tender documents.
- Failure to submit UNPRICED bid format with their OEM and Product details such as part codes.
- Bids with terms and conditions that cannot be accepted by DOT / buyer.
- If the information provided by the Bidder is found to be incorrect / misleading / fraudulent at any stage / time during the evaluation process.
- If the bidder proposed OEM is unable to perform the PoC for a particular criterion, where the OEM has declared to have the feature / functionality – in which case, the bidder(s) of the respective OEM shall be disqualified / rejected from further evaluation. Also the OEM shall be liable for black listing by DOT.

Commercial Rejection Criteria

- Incomplete price bid
- Any attempts to alter our promulgated price bid format or Bids including quantity that do not conform to the tender's price bid format.
- Overwriting or white ink corrections if any
- Price bids with bidder's own terms and conditions
- Any attempts to mislead the evaluators or bids with blank GST or levies as applicable.

- If the information provided by the Bidder is found to be incorrect / misleading / fraudulent at any stage / time during the evaluation process.

5.3. Confidentiality of the Document

The Bidder will treat all data & information about DOT, obtained in the execution of its responsibilities as confidential & will not reveal such information to any other party without prior written approval of DOT. If the Bidder leaks any such information to any third party by any means, DOT holds the right to take such action as may be necessary.

Disclosure of any part of information to parties not directly involved in providing the services requested could result in the disqualification of the Bidder, pre-mature termination of the contract and/or legal action against the Bidder for breach of trust.

No news release, public announcement, or any other reference to this RFP or any program there under shall be made without written consent from DOT. Reproduction of this RFP, without prior written consent of DOT, by photographic, electronic, or other means is prohibited.

Conflict of Interest

Bidder(s) must disclose to DOT in their proposal any potential conflict of interest, including any conflict which may involve DOT employees who may have a financial interest in the Bidder. If such conflict of interest exist, DOT may, at its discretion, refuse to consider the Proposal.

Debarment

In case of any misconduct or fraudulent practice, bidder may be debarred.

Assignment

The Bidder shall not assign, in whole or in part, its obligation to perform under this contract, except with DOT prior written consent. The Bidder shall notify DOT in writing of all sub- contracts awarded under the contract, if not already specified in his bid. Such notification, in his original bid or later, shall not relieve the Bidder from any liability or obligation under the contract.

Good Faith Statement

All information provided by DOT in this RFP is offered in good faith. Individual items are subject to change at any time. DOT makes no certification that any item is without error. DOT is not responsible or liable for any resulting claims arising out of use of this information.

5.4. NDA (Non-Disclosure Agreement)

The successful H1 bidder must sign the 'Non-Disclosure Agreement (NDA)' on Rs. 200/- stamp paper (non-judicial) from their competent authority as a compliance for the 'Non-Disclosure Agreement' in line with DOT IS Security Policy. Purchase orders will not be placed without entering into above NDA. In the event of failure on the part of the successful Bidder to sign the agreement in the above- stipulated period, the DOT may cancel the order.

5.5. Termination Clause

DOT at its absolute discretion, reserves its right to terminate the contract/ agreement for any reason including but not limited to the following:

- DOT without prejudice to any other remedy for breach of contract, may terminate the contract/agreement by giving seven days' notice in the event of unsatisfactory performance or on breach of any stipulated conditions or qualitative dimensions of the various services specified/agreed upon by the selected bidder, or if the engagement is not in the interest of DOT or DOT no more requires any such service.
- Other Grounds for Termination: DOT is entitled to terminate this contract/agreement for any reason at its absolute discretion forthwith without notice, without assigning any reason and without payment of any compensation, in the following cases: -
- The Bidder is adjudicated insolvent by a Competent Court or files for insolvency.
- It is clearly understood by the Bidder that if a charge sheet is filed by any competent authority of the Government against the Bidder, the Bidder is obliged to notify DOT within fifteen days of filing of the charge sheet. Failure to do so shall result in forfeiture of all payments due for service rendered after the date of the filing of the charge sheet.
- For any reason whatsoever, the selected bidder becomes disentitled in law to perform his obligations under this contract/agreement.
- The bidder is involved in wrongful billing. In addition, hereto wrongful billing shall also result in the bidder being debarred in accordance with guidelines of DOT.

5.6. Resolution of Disputes

- All disputes and differences of any kind whatsoever arising out of or in connection with the contract, whether during the progress of the works or after their completion shall be referred by the Successful bidder to owner and owner shall within a reasonable time after their presentation make and notify decisions thereon in writing.
- If dispute or difference of any kind shall arise between the DOT and the successful bidder in connection with or relating to the contract, the parties shall make every effort to resolve the same amicably by mutual consultations.
- If the parties fail to resolve their dispute or difference by such mutual consultation within twenty-one days of its occurrence, then, unless otherwise provided in the tender document, either the DOT or the successful bidder may give notice to the other party of its intention to commence arbitration, as provided the applicable arbitration procedure will be as per the Arbitration and Conciliation Act, 1996 of India.
- In the case of a dispute or difference arising between the DOT and a domestic Successful bidder relating to any matter arising out of or connected with the contract, such dispute or difference shall be referred to the sole arbitration of the Board of Directors whose decision shall be final.
- Venue of Arbitration: The venue of arbitration shall be the place from where the contract has been issued, i.e., Trivandrum, Kerala State, India.
- The contract shall be governed by and interpreted in accordance with the laws of India for the time being in force.
- All disputes arising out of this tender will be subject to the jurisdiction of courts of law in Trivandrum

Indemnity

The bidder shall indemnify to protect and save DOT, its employees, personnel, officers, directors and representatives against all claims, losses, costs, damages, expenses, action suits and other proceedings.

Violation of Terms

DOT clarifies that DOT shall be entitled to an injunction, restraining order, right for recovery, suit for specific performance or such other equitable relief as a court of competent jurisdiction may deem necessary or appropriate to restrain the Bidder(s) and its Partner(s) from committing any violation or enforce the performance of the covenants, obligations and representations contained in this RFP. These injunctive remedies are cumulative and are in addition to any other rights and remedies. DOT may have at law or in equity, including without limitation, a right for recovery of any amounts and related costs and a right for damages.

Merger/ Acquisition of Bidder

In the event of the Bidder's Firm or the concerned division of the firm being taken over/bought over by another firm, all the obligations under the agreement with DOT should be passed on for compliance to the new firm in the Negotiations for their transfer.

5.7. Delays in the Bidder's Performance

If at any time during performance of the Contract, the Bidder should encounter conditions impeding timely performance of services, the Bidder shall promptly notify DOT in writing of the fact of the delay, its likely duration and its cause(s).

DOT reserves the right to reject a bidder in case it is observed that they may not be in position to execute this job as per the required schedule. The decision of DOT will be final in the regard.

As soon as practicable after receipt of the Bidder's notice, DOT shall evaluate the situation and may at its discretion extend the Bidder's time for performance.

Preliminary Examinations

- DOT will examine the Bids to determine whether they are complete, the documents have been properly signed, supporting papers/documents attached and the bids are generally in order.
- DOT at its sole discretion, may waive any minor nonconformity or irregularity in a Bid which does not constitute a material deviation, provided such a waiver does not prejudice or affect the relative ranking of any Bidder.
- Prior to the detailed evaluation, DOT will determine the substantial responsiveness of each Bid to the Bidding document. For purposes of these Clauses, a substantially responsive Bid is one, which conforms to all the terms and conditions of the Bidding Document without material deviations.
- If a Bid is not substantially responsive, it will be rejected by DOT and may not subsequently be made responsive by the Bidder by correction of the nonconformity.

Bidder(s) are expected to examine all instructions, forms, terms and specification in this RFP. Failure to furnish all information required by this RFP or to submit a Bid not substantially responsive in every respect will be at the Bidder's risk and may result in the rejection of Bid.

Withdrawal of Bids

No bid may be withdrawn in the interval between the last date for receipt of bids and the expiry of the bid validity period specified in this RFP document.

Transition Process

Upon termination or expiration of this Contract, DOT and the Bidder shall reasonably cooperate with each other to affect a smooth transition so as not to impose undue hardship.

5.8. Corrupt or Fraudulent Practices

- It is required by all concerned to observe the highest standard of ethics during the procurement process. In pursuance of this policy, the DOT prescribes the following conditions:
- "Corrupt practice" means the offering, giving, receiving or soliciting of anything of value to influence in the procurement process or in contract execution; and "fraudulent practice" means a misrepresentation of facts in order to influence a procurement process or the execution of a contract to the detriment of the DOT, and includes collusive practice among Bidders (prior to or after Tender submission) designed to establish Tender prices at artificial non-competitive levels and to deprive the DOT of the benefits of free and open competition;
- DOT will reject a proposal for award if it determines that the Bidder recommended for award has engaged in corrupt or fraudulent practices in competing for the contract in question; will declare a firm ineligible, either indefinitely or for a stated period of time, to be awarded a contract by the DOT if it at any time determines that the firm has engaged in corrupt or fraudulent practices in competing for, or in executing the contract.
- No bidder shall contact the DOT or any of its officers on any matter relating to its bid, other than communications for clarifications and requirements under this tender in writing, with an intention to influence the members of various committees or officials of DOT. Any such effort by a bidder to influence the DOT in the DOT's bid evaluation committee, bid comparison or contract award decisions may result in rejection of the bidders bid.

5.9. Force Majeure

- For purposes of this clause, Force Majeure means an event beyond the control of the successful bidder and not involving the successful bidder's fault or negligence and which is not foreseeable and not brought about at the instance of the party claiming to be affected by such event and which has caused the non–performance or delay in performance. Such events may include, but are not restricted to, acts, wars or revolutions, hostility, acts of public enemy, civil commotion, sabotage, fires, floods, explosions, epidemics, quarantine restrictions, strikes excluding by its employees, lockouts excluding by its management, and freight embargoes.
- If a Force Majeure situation arises, the successful bidder shall promptly notify the DOT in writing of such conditions and the cause thereof within twenty-one days of occurrence of such event. Unless otherwise directed by the DOT in writing, the successful bidder shall continue to perform its obligations under the contract as far as reasonably practical and shall seek all reasonable alternative means for performance not prevented by the Force Majeure event.
- If the performance in whole or in part or any obligation under this contract is prevented or delayed by any reason of Force Majeure for a period exceeding sixty days, either party may at its option terminate the contract without any financial repercussion on either side.
- In case due to a Force Majeure event the DOT is unable to fulfil its contractual commitment and responsibility, the DOT will notify the successful bidder accordingly and subsequent actions taken on similar lines described in the above sub-paragraphs.

5.10. Service Level Agreement

This SLA defines the agreed-upon service levels, performance metrics, and responsibilities for the SD-WAN solution provided by the bidder. The SLA Section outlines the expectations for SLA Metric, Penalties in case of SLA breaches and Incident Management and Response Times

5.10.1. Service Level Agreement Metric

5.10.1.1 DC and DR Appliances

- Availability Service Level will be measured on a monthly basis for SD-WAN Centralized Network Orchestrator/Manager/Controller Devices and Head End Devices.
- Available Time will be calculated for 24 Hours
- SLA report needs to be submitted on quarterly basis within 15 days of the beginning of next Quarter.

The Services Availability shall be calculated on monthly basis as follows:

$$\text{Availability} = \frac{(\text{Total No. of hours in a month} - \text{Total Outage Hours in Month})}{(\text{Total No. of Hours in a Month})} \times 100$$

Total No. of Hours in a Month= No. of working days x 24 hours

Sl.No.	Uptime Per device *Note: Per device uptime penalty shall be levied even if redundant device/devices installed in pair (High Availability) is working fine and there is no impact on the application traffic.*	Penalty per Device for DC and DR
1	99.5% and above per device	Nil
2	If uptime is <99.5% and up to 99.3%,	Rs.3,000/-
3	If uptime is <99.3% and up to 99.1%,	Rs.5,000/-
4	If uptime is <99.1% and up to 99.0%,	Rs.8,000/-
5	If uptime is <99.0% and below	Rs.15,000/-

5.10.1.2 Branch Devices

- Availability Service Level will be measured on a monthly basis for SD-WAN Individual Branch Devices Type1 and Type2
- Available Time will be calculated between 8 AM to 8 PM for the Day. (12 Hours)
- SLA report for appliances needs to be submitted on quarterly basis within 15 days of the beginning of next Quarter.

The Services Availability shall be calculated on monthly basis as follows:

$$\text{Availability} = \frac{(\text{Total No. of hours in a month} - \text{Total Outage Hours in Month})}{(\text{Total No. of Hours in a Month})} \times 100$$

Total No. of Hours in a Month = No. of working days x 12 hours

Sl. No.	Uptime Per device *Note: Per device uptime penalty shall be levied even if redundant device/devices installed in pair (High Availability) is working fine and there is no impact on the application traffic.*	Penalty per Device for Branches (TYPE1 and TYPE2
1	99.5% and above per device	Nil
2	If uptime is <99.5% and up to 99.3%,	Rs.1,000/-
3	If uptime is <99.3% and up to 99.1%,	Rs.3,000/-
4	If uptime is <99.1% and up to 99.0%,	Rs.5,000/-
5	If uptime is <99.0% and below	Rs.10,000/-

Note:
Bidder will submit the down time report along with the invoice.
The downtime calculated shall not include the following.

- Loss of power at the location or instable power source.
- Location is unable to conduct business due to an event beyond the control of bidder that may affect the SD-WAN service (i.e., act of God, weather related events, fire, civil unrest, terrorist attack)
- Any service interruption caused by any non-SD-WAN provided equipment or service e.g. downtime due to scheduled preventive maintenance or routine servicing etc. is excluded.
- End-of-Life Components: Issues arising from using components beyond their recommended operational lifespan are not considered inherent failures.
- Failure or malfunction of any equipment or services not provided by the Bidder.

5.10.2. Incident Management and Response Time

Bidder must ensure that solution meets the agreed business requirement. In case of services failure, Bidder may be liable for penalties. DOT may also be entitled to terminate the contract without any liability or obligation. Successful bidder should stick to the following turnaround time:

Incidents will be categorized as follows:

Sl. No.	Severity	Severity Level	Initial Response Time	Temporary Resolution	Resolution Time
1	High	P1	30 Minutes	1 Hour	4 Working Hours
2	Medium	P2	1 Hour	2 Hours	8 Working Hours
3	Low	P3	4 Hours	12 Hours	24 Working hours

The Severity Levels are categorised as below:

Sl. No.	Severity	Severity Level	Details
1	High	P1	Complete service outage or severe degradation impacting multiple applications or users.
2	Medium	P2	Partial service degradation affecting critical applications or users.
3	Low	P3	Minimal impact, including non-urgent requests or queries.

In case of the bidder penalised at highest penalty for 3 consecutive quarters, The DoT may consider immediate cancellation of the contract and the remaining payment for the contract will be forfeited. The support for the remaining contract period will be the responsibility of the OEM.

5.11. Training

- The OEMs of SD-WAN solution and Network Monitoring cum Helpdesk Platformshould provide On-Site Training to the DOT's nominated IT Team Members. The training should include but not limited to following modules.
    - Overview to SD-WAN solution and Network Monitoring cum Helpdesk Platform
    - Design, Deployment, Configuration and Reporting
    - Troubleshooting and Maintenance

5.12. On-site Technical Resource requirement

5.12.1. SD-WAN and Network Infrastructure Management

1. The Bidder should maintain the SD WAN solution and Network Infrastructure for a Period of 1 Year from the date of Sign of the Project.
2. The Bidder to deploy Technical Engineer for management and maintenance of the Network Infrastructure (Switches)and SD-WAN solution at Trivandrum Head Office for a period of 1 Year. The Working Hours will be as per DOT's official timings. The Technical Engineer will support the Network Infrastructure and SDWAN solution implemented at DOT. The Technical Engineer will report to DOT's IT Chief Co-ordinator.
3. The Bidder must ensure the following with respect to qualification, experience and background check of the Technical Engineer:

a) Qualification:

The Technical Engineer should hold a Diploma or B. E/ B. Tech Degree.

b) Experience:

The Technical Engineers should have minimum 4 years of hands-on experience in administration of the Network Infrastructure and SD-WAN Solution.

c) Background check:

Successful bidder must conduct proper background check of the Technical Engineer and enter into NDA with DOT.

d) Duties pertaining to the Network Infrastructure and SD-WAN Solution but not limited to.

Overall operations of the Network Infrastructure and SDWAN solution which includes Monitoring, Management, Configuration or any Changes/Modifications, Troubleshooting, Reporting, SW Updates to be done for as per the business requirement of DOT.

- Management and monitor the availability of links and their utilization and take corrective action where required.
- Monitor the critical errors in SD-WAN and Network Infrastructure and taking actions.
- Maintain a record of Hardware and software configurations of all equipment including the details of different policies implemented on the Network and SD-WAN devices.
- Keep regular backups of soft copies of the configurations of each of the devices
- Provide services to different users on demand basis as and when required as.
- Proactive monitoring of health of the Network Infrastructure and SD-WAN solution, including the H/W and S/Won various parameters such as CPU, Memory, Interface Utilizations, etc and Report abnormalities to the DOT as and when observed/occurred.
- Troubleshooting Day to day issues pertaining to Network Infrastructure and SDWAN solution in coordination with DOT's Network integrator, or other relevant teams/vendors.
- Call logging and follow-up with the OEM and escalation for resolution of of Hardware and Software issues.
- Preparing and submitting reports as per the requirement of the DOT.
- The Technical Engineer must ensure the Regular back up of configuration and data are in place.

5.12.2. Network Monitoring cum Helpdesk Platform

1. The Bidder should maintain the Network Monitoring cum Helpdesk Platform for a period of 1 Year from the date of Sign of the Project.
2. The Bidder to deploy Technical Engineer for management and maintenance of Network Monitoring cum Helpdesk Platform at Trivandrum Head Office. The Working Hours will be as per DOT's official timings. The Technical Engineer will support Day-Today Operation of Network Monitoring cum Helpdesk Platform implemented at DOT. The Technical Engineer will report to DOT's IT Chief Co-ordinator.

3. The Bidder must ensure the following with respect to qualification, experience and background check of the Technical Engineer:

a) Qualification:

The Technical Engineer should hold a Diploma or B. E/ B. Tech Degree.

b) Experience:

The Technical Engineers should have minimum 4 years of hands-on experience in administration of the Network Monitoring cum Helpdesk Platform.

c) Background check and NDA

Successful bidder must conduct proper background check of the Technical Engineer and enter into NDA with DOT.

d) Duties pertaining to the Network Monitoring cum Helpdesk Platform

- Overall management of the Network Monitoring cum Helpdesk Platform which includes Management, Configuration or any changes/modifications, SW Updates to be done for as per the business requirement of DOT.
- Proactive Monitoring of health of the IT infrastructure (Network Devices, End User Systems and Service Provider Links) and Report abnormalities to the DOT as and when observed/occurred.
- Work closely with IT Operations Team and End Users of DOT to Manage Tickets opened via Helpdesk tool w.r.t Ticket creation, assignment, escalation, SLA, Resolution and Closure.
- Call logging and follow-up with the OEM and escalation for Software issues
- Preparing and submitting reports as per the requirement of the DOT. Reports will include daily health monitoring and other statistical reports. The Technical Engineer may take support from its Backend team and/or OEM if required.
- The Technical Engineer must ensure the Regular back up of configuration and data are in place.

6. Tender Forms

Sl. No.	Item Description	Annexure No.
1	Bidder's General Information & Eligibility Criteria	Annexure 1
2	Declaration by Authorised Signatory of Bidder	Annexure 2
3	Bidder Turn Over details	Annexure 3
4	Non-Blacklisted Declaration - Bidder & OEM.	Annexure 4
5	MAF (OEM Authorization)	Annexure 5
6	No Deviation Certificate	Annexure 6
7	QCBS Evaluation	Annexure 7 (A)
	Technical Specifications SD-WAN	Annexure 7 (B)
	Technical Specifications Network Monitoring cum Helpdesk Platform.	Annexure 7 (C)
	Technical Specifications Desktop	Annexure 7 (D)
	Technical Specifications Professional Display Monitor	Annexure 7 (E)
8	Escalation Matrix - Bidder & OEM	Annexure 8
9	Agreement	Annexure 9
10	Bidder & OEM Purchase Order Details	Annexure 10
11	Unpriced Bill of Material	Annexure 11
12	End of Sale /End of Support	Annexure 12
13	Price Breakup	Annexure 13
14	Self-Declaration for QCBS Technical points	Annexure 14

Annexure 1- Bidder's General Information & Eligibility Criteria

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: Bidder's General Information & Eligibility Criteria
Tender Reference No:

Sl. No.	Particulars (Information Required)	Bidder's response	Document Page Nos.
1	Name of the Company		
2	Address of the Firm / Company		
3	Year of Registration/Incorporation		
4	Name & Telephone Number of the Proprietor/person to whom all references shall be made regarding tender:		
	• Name of the person		
	• Telephone No (Landline)		
	• Mobile No.		
	• E-mail Id		
5	Details of the Bank Account of the Bidder		
	• Name of the Bank		
	• Branch and address		
	• IFSC Code		
6	Turnover of the bidder for last 3 individual years (2021-22, 2022-23, 2023-24) should be Rs. 10 Crore or above. (Relevant pages of the Audited Financial Statement shall be uploaded).		
	Audit Balance Sheets of last three FY viz. FY 2021-22, FY 2022-23 & FY 2023-24.		
	CA Certificate indicating the turnover for the last 3 years, and Positive Net worth as on 31 March 2024. after duly evaluating the financial strengths vis-à-vis the technical expertise of the company.		
7	Bidder should have experience and expertise of having successfully completed any IT Network projects of the following value, in addition to 1 SD-WAN Purchase Order, Any 1 Purchase Order of value 7 Crores or above, (Or) 2 Purchase orders of value 4 Crores or above (Or) 3 Purchase orders of value 3 crores or above. **Note: All Purchase Orders must be from an organisation under Govt of India.**		

8	The bidder(s) should not be blacklisted or banned for business by any Public Sector undertaking / Govt. Department/ MNC/ Private Limited Companies in India or abroad.		
9	Key Personnel of the Company:		
10	• Name		
	• Designation		
	• Email ID		
	• Contact Number		
	• Acceptance of all terms and conditions mentioned in the tender Document by Bidder(s).		
11	Acceptance of all terms and conditions mentioned in the tender Document by Bidder(s).		

Note: Bidder(s) should submit all the required documents as per eligibility Criteria and should also mention all the page numbers of supporting documents of Eligibility Criteria.

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal]

Annexure 2- Declaration by Authorised Signatory of Bidder

Note for Bidder:

- To be submitted on Bidder's letter head

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: Declaration by Authorised Signatory of Bidder
Tender Reference No:

I/We hereby certify that all the information and data furnished by me/us with regard to the above Tender Specification are true and complete to the best of my/our knowledge. I/We have gone through the specifications, condition, stipulations and other pertinent issues till date, and agree to comply with all the tender requirements including 3-year CAMC and Intent of the specification.

I/We further certify that I/We am/are authorized to represent on behalf of my/our firm/company for the abovementioned tender and a valid Power of Attorney/Authorization letter to this effect is also enclosed.

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal]

Annexure 3 - Bidder Turnover details

Note for Bidder:

- To be submitted on Bidder's letter head

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: Bidder Turnover details
Tender Reference No:

Sl. No.	Financial Year	Total Turn Over Value as per Audited Statement	Profit Value as per Audited Statement
1	FY.2021-22		
2	FY.2022-23		
3	FY.2023-24		

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal]

Annexure 4- Non-Blacklisted Declaration - Bidder & OEM.

Note for Bidder and OEM

- To be submitted on Bidder's and OEM's letter head along with Technical Bid

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: Non-Blacklisted Declaration

Tender Reference No:

I/We confirm that my/our firm/company is not blacklisted in any manner whatsoever by any central Government department, autonomous organizations, Public Sector, Private Sector, Undertakings (PSUs) or any other Government Organizations in India on any ground including but not limited to indulgence in corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practice.

It is hereby confirmed that I/We are entitled to act on behalf of our firm/company and empowered to sign this document as well as such other documents, which may be required in this connection.

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal

Note: In absence of above declaration / certification, the Bid is liable to be rejected.

Annexure 5 - MAF (OEM Authorization)

Note to the OEM:

- This letter of Authorization should be on the Letterhead of the Manufacturer by the OEM Authorized Signatory only.

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: Manufacturer Authorization Form for Tender Ref# (Tender reference Number)
Tender Reference No:

We, <OEM/Manufacturer name>, having our registered office at <OEM/Manufacturer Address> are a reputable manufacturer of <OEM/Manufacturer Product Line offered>.

We hereby confirm that <Bidder Name> with the registered address at <Bidder Address> is our registered partner and we hereby authorize <Bidder Name> to participate and execute the order for subject tender on behalf of <OEM/Manufacturer name>

Our full support is extended to them in all respects for Supply, Installation & Maintenance of our products. We also ensure to provide Back-to-Back service support for the supplied solution as per the RFP terms & conditions during the entire warranty period of 5 years from the date of Supply / Installation of the solution.

We also undertake that in case of default in execution of this project tender by the <Bidder Name>, the <OEM/Manufacturer Name> will take all necessary steps for successful execution of this project as per tender requirements.

Thanking You,
For<OEM/Manufacturer Name>
<Authorized Signatory>
Name:
Designation:
Date:
Place:
[Company Seal]

Annexure 6 - No Deviation Certificate

Note to the Bidder:

- To be submitted on Bidder's letter head

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: No Deviation Certificate
Tender Reference No:

I/We hereby confirm that I/We have not changed/modified/materially altered any of the tender documents received from The Directorate, Directorate of Treasuries, office and in case of such observance at any stage, it shall be treated as null and void and my/our tender shall be deemed to be withdrawn.

I/We also hereby confirm that I/We have neither set any Terms and Conditions and nor have I/We taken any deviation from the Tender conditions together with other references applicable for the above referred RFP/Tender Specification.

I/We further confirm my/our unqualified acceptance to all Terms and conditions, unqualified compliance to Tender Conditions, Integrity Pact, and acceptance to bidding process.

I/We confirm to have submitted offer in accordance with RFP instructions and as per aforesaid reference.

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal]

Annexure 7 (A) - Declaration of Qualification for Quality Points mentioned in QCBS

Note for Bidder's:

- To be submitted on Bidder's Letterhead on behalf of the OEM's

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: Declaration of Qualification for Quality points mentioned in QCBS
Tender Reference No:

S.No	Category	Max. Marks	Description	Marks	Document to be submitted
1	SDWAN OEM Experience Government Department/BFSI/ PSU/PSE/Private sector	10	SDWAN Experience in India (As on Bid calling Date) a) More than 10 years - 5 Marks b) More than 7.5 years - 3 Marks c) More than 5 years - 2 Marks	5	PO copy with Date clearly mentioned
2			SDWAN Experience in India a) More than 500 location Project - 5 Marks b) More than 250 multisite Project - 3 Marks c) Less than 100 multisite Project - 2 Marks	5	PO copies
3	SD-WAN Technical Specifications (Core functionalities)	35	Capability to integrate different types of connectivity (MPLS/ILL/FTTH/Broadband/SIM/VSAT) into a single, end-to-end IP Framework and IP tunnel that uses policies to manage traffic.	5	Data Sheet and PoC
4			Capability to integrate different types of connectivity for Active- Active Load Balancing without traffic duplication on the connected Links at a site (MPLS/ILL/FTTH/Broadband/SIM/VSAT)	15	Data Sheet and PoC
5			Capability for DNS based Inbound Load Balancing of Multiple divergent media	5	Data Sheet and PoC
6			Capability of Seamless Failover of Voice/Video and Data Traffic less than 20 Milliseconds	5	Data Sheet and PoC
7			Capability for switching traffic to new DHCP IP's from Service Providers for multiple DHCP based Links at sites	5	Data Sheet and PoC
8	SD-WAN Technical Specifications (Security)	15	Capability of inbuilt Firewall and IPS in DC/DR/Branch Devices	5	Data Sheet and PoC
9			Capability of Selective Encryption of Links at sites	5	Data Sheet and PoC
10			Traffic Encryption capability shall be FIPS 140-2 Compliant	5	Data Sheet and PoC
11	SD-WAN Technical Specifications	5	Capability to Prioritise and steer traffic based on Applications	5	Data Sheet and PoC

	(Application Awareness)				
12	SD-WAN Technical Specifications (WAN Optimization)	10	Built In Advanced WAN Optimisation Capabilities - Caching, Compression, Deduplication of Application traffic	5	Data Sheet and PoC
13			Capability of WAN Optimisation at Byte Level	5	Data Sheet and PoC
14	The proposed Network Monitoring cum Helpdesk Platform OEM Experience	5	Network Monitoring cum Helpdesk Platform Experience in India - (in state/central Government/ PSU) a) Minimum 3 Projects with 10000 Devices - 5 Marks b) 2 Projects with 10000 Devices - 3 Marks c) 1 Project with 10000 Devices - 1 Marks	5	Reference PO copy and completion/ Signoff Documents need to be submitted.
15	The proposed Network Monitoring cum Helpdesk Platform OEM certifications	5	a) The Proposed Network Monitoring and Helpdesk Platform OEM should be OWASP & SANS certified from cert-in empanelled vendor. b) The Proposed Network Monitoring cum Helpdesk Platform OEM should be certified for ITIL V4 for 9 or more practices. c) The proposed Network Monitoring cum Helpdesk Platform OEM should have ISO 270001, ISO 27034, SOC3 and CMMI Level 3	5	Documents need to be submitted for all 3 points
16	Presentation by Technically Qualified OEM's	15	Each SD-WAN and Network Monitoring cum Helpdesk Platform OEM's will be allotted 30 minutes time slot for presenting (ON-LINE) the technical capabilities, features, proposed components, and deployment approach in line with RFP requirements. SD-WAN OEM - 10 Marks Network Monitoring cum Helpdesk Platform OEM - 05 Marks	15	Presentation to Technical Committee
	TOTAL	100		100	

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal]

Annexure 7 (B) - Technical Specifications SD-WAN

Note: Following Technical Specifications compliances to be submitted on OEM Letterhead

- Functional Requirements
- Centralized Network Orchestrator/Manager/Controller
- Head-end Devices
- Branch Devices Type-1: District Treasury Office
- Branch Devices Type-2: Sub Treasury Office

Technical Specification – Functional Requirement		
Sr. No	**Requirements**	**Compliance Yes /No**
Architecture, Deployment and Performance		
1	The proposed SDWAN Solution should be implemented as true software defined WAN solution. The proposed solution should have Central Management solution/Controller/Orchestrator, Head-End and Branch Devices.	
2	The proposed SD-WAN Solution shall be implemented as secure software-defined network Architecture with a complete separation of Control, Data and Management plane.	
3	The proposed SD-WAN Solution shall integrate different types of connectivity (MPLS, ILL, FTTH, Broadband, SIM, ADSL/DSL, Ethernet, Fibre, LTE/3G/4G/5G, satellite) into a single, end-to-end framework and encrypted tunnel that uses policies to manage traffic.	
4	The communication between the central location and the branch device running on the remote entity should be secure and encrypted.	
5	The proposed SD-WAN solution must include following components in a OEM Physical Appliance Form Factor: 1. Central Management solution/Controller/Orchestrator 2. Central SD-WAN Head-End appliances 3. Branch SD-WAN appliances	
6	The proposed SD-WAN Central Management solution/Controller/Orchestrator need to be installed on-premises	
7	The proposed SDWAN solution must co-exist with existing IP/MPLS network	
8	The proposed Central Management solution/Controller/Orchestrator solution should be a single application and not combination multiple application that expected to deliver that provides global visibility and control over multiple SD-WAN appliance. It must streamline management tasks such as device configuration, policy deployment, across the network. SDWAN Management solution should provide single platform to monitor, troubleshoot and maintain the SDWAN Features.	
9	The proposed SD-WAN solution must have separation of data and management to achieve following functionality:	

	Central Management solution/Controller/Orchestrator solution should be deployed in HA at DC and DR. a. Central Management solution/Controller/Orchestrator to provide single point of entry for Configuration and Monitoring. Central Management solution/Controller/Orchestrator shall be securely accessed and shall be capable of configuration, monitoring and troubleshooting of SD-WAN devices in the branches and Datacentres. All the devices be managed from the Central Management solution/Controller/Orchestrator. Central Management solution/Controller/Orchestrator shall maintain centralized routing table, controls route advertisement as per policy, creates end to end segments on network, instructs data plane to change traffic flow as per policy.	
	b. Data Plane: Forwards traffic in encrypted tunnels, applies local policy like QoS and security policies.	
	c. Control Plane: Maintains centralized routing table, controls route advertisement as per policy, creates end to end segments on network, instructs data plane to change traffic flow as per policy.	
	d. SD-WAN appliances: SD-WAN Head End Devices: The Head Devices in DC and DR will be deployed in HA mode. SD-WAN Branch Device: All the Branch SD-WAN devices will be deployed in Standalone Mode.	
10	All SD-WAN solution components including the Central Management solution/Controller/Orchestrator and Edge SDWAN Devices shall be from the same OEM	
11	The Central Management Solution/Controller/Orchestrator shall be designed to support up to 2000 devices and any required SDWAN licenses for controllers shall be provided from day-1.	
12	From the Central Management Solution/Controller/Orchestrator DOT should be able to configure Template for branch and central locations	
13	The proposed SD-WAN Solution shall support deployment of any kind of topologies such as full-mesh, hub-spoke and partial mesh as desired by DOT. It shall be possible to change network topology only by pushing policy on Central Management Solution/Controller/Orchestrator and shall not require device by device configuration. The proposed SDWAN solution should allow multiple Hub Sites (at least 5) when deployed in multiple hubs and spoke topology.	
14	The solution shall support Customized Role Based Access Control that provides only relevant information to the user based on their roles and privileges. Solution shall also provide customized role builder and support multiple administrators. Solution shall provide detailed audit trail of changes etc. The Proposed SDWAN reporting and dashboard solution should have capacity for minimum 10 concurrent login for the admin/management console or GUI without any degradation in services.	
15	The Central Management Solution/Controller/Orchestrator shall support local, external Radius Server User/Account authentication method or two factor authentications and AD integration.	

16	The proposed SDWAN solution should have the capability to do forward error correction to mitigate against packet drops automatically when SLA (packet loss, latency, and jitter) of available link(s) is/are violated for defined application/traffic.	
17	In case of failure of Central Management Solution/Controller/Orchestrator the remote device should work seamlessly and should be accessed via GUI.	
18	The proposed solution shall allow for plug and play installation of Branch devices without requiring any manual configuration at the remote location.	
19	All elements of the solution will have IPv4 and IPv6 support from Day1	
20	The SDWAN Central Management Solution/Controller/Orchestrator shall have Capability for Zero Touch Provisioning. There shall be “No impact” on Data Plane traffic in case of Controller/Orchestrator outage.	
21	The system should be able to select the optimum path based on the network parameters like Latency, Jitter, PLR and network capacity.	
22	Solution should able Granular application and service controls allow DOT to prioritize applications based on whether the application is mission-critical, latency-sensitive, or meets certain health criteria	
23	The system should be able to prioritize business critical applications and should prioritize this traffic over others during congestion	
24	The system should be able to automatically steer traffic flows to the optimum path, based on policy definition in the software defined network controller	
Dynamic Load Sharing of Traffic Based on Business and/or application policies		
25	The proposed SDWAN Head-End and Branch devices shall be able to load balance traffic across multiple connectivity paths based on load balancing algorithms efficiently using all available connectivity. System shall also detect blackouts & brownouts by supporting active-active load balancing and fast session failover. Shall be able to integrate different types of connectivity (MPLS/ILL/FTTH/Broadband/SIM/VSAT) for Active-Active Load Balancing on per packet, per session and per Application basis with Seamless Failover without traffic duplication on the connected links at a site.	
26	The proposed Edge devices shall be able to identify and classify applications at OSI layer L4-L7.	
27	The proposed solution shall be able to dynamically control data packet forwarding decisions by looking at application type, performance, policies, and path status	
28	The proposed solution shall allow various traffic load balancing requirement like, between equal and unequal bandwidth links, application flow via specific link only.	
29	The SDWAN shall be able capable of Seamless Failover of Voice/Video and Data Traffic	
30	The system shall do continuous performance monitoring of latency for all network paths, links and make decisions to forward critical applications over the best performing path based on the defined application policy.	

31	The SD-WAN solution shall improve the degraded link (e.g. Forward Error Correction/SACK or equivalent, Quality of Service (QoS), Jitter buffering etc.) to maintain application performance for high priority applications	
32	In the event of link degradation or link down the solution shall be able to move traffic to a better link without any application disruption by moving the data traffic on a per packet/flow basis	
33	The proposed solution shall be able to automatically route over the overlay WAN without the need for a separate routing protocol. Solution shall support automatic overlay routing without any manual configuration.	
34	The proposed solution shall be able to prioritize business critical applications and shall have the capability of prioritizing traffic during congestion	
35	The proposed solution shall make path optimization decisions on the Edge device and not through the central controller to reduce delay and improve performance.	
36	Proposed SDWAN solution should have capability to define Application Aware Routing mandatorily in all the following manners:	
	a. Traffic for any application / group of applications should automatically select the best path, based on the real-time system calculated parameters like Latency, Jitter, Packet loss etc. The application traffic must switchover to alternate link/path even if any one of the configured parameters violate the predefined SLA, irrespective of parameter priority	
	b. There should be provision to bind an application / Group of applications to a specific link. However, the traffic bounded to a specific link should auto switchover on alternate available link in cases like Link down, Link violates any of the configured SLA, irrespective of parameter priority.	
	c. There should be option to distribute application's traffic to all the available links, for which link parameters are within defined	
37	In the proposed SDWAN solution, Head-End and Branch device should be able to monitor different path parameters for all the WAN Links in near real time.	
38	In the proposed SDWAN solution, the WAN path selection for the traffic originating from branch should be dynamically done based on the policy configured/pushed from the Central Management Solution/Controller/Orchestrator.	
39	The device should support load balancing on available WAN links based on	
	1. Session based	
	2. Packet based	
40	The solution should be able to load balance traffic among various links without traffic duplication on links and the solution shall seamlessly failover traffic on failure of primary link to secondary and tertiary links in sub second interval. The failover shall seamlessly work among various type of connectivity's e.g. MPLS / Internet / Broadband /4G LTE / 5G /VSAT etc. The seamless failure implies that sessions should not break while switching	

	traffic from Primary to other links and vice versa.	
41	The Proposed SDWAN solution should load balance the critical application (Highest Priority) traffic through second best link and so on, if the best link bandwidth is utilized up to configured threshold.	
42	The proposed SDWAN solution should be able to load balance the traffic across multiple WAN links of uneven bandwidth. Simultaneous monitoring of the links should be done to avoid congestion on the lower-bandwidth link.	
43	The proposed SDWAN Solution should be able to utilize all types of service provider links irrespective of the last mile connectivity type (e.g., MPLS, 4G, VSAT, RF, 5G, etc.) simultaneously and load balance the traffic without traffic duplication amongst the all the links irrespective of the last mile connectivity as per configured policies. The devices should support Ethernet extender to support 4G and 5G connectivity (4G, 5G devices will be provided by Telecom Service Provider and Ethernet cable will be extended till router).	
44	The proposed SDWAN solution should continuously monitor the link flaps. If the link is not stable, then the traffic to be diverted on alternate superior link. The inferior link to be put in monitor state with a configurable time interval. Once, the inferior link is stable and within the SLA parameters configured, all existing as well as new sessions must shift to superior link accordingly as per configured policy.	
45	The proposed SDWAN solution should support seamless application accessibility across Datacentres & Branches during auto failover of WAN links and load balancing without any session disconnection and traffic duplication. The session switches over or switch back between links without any manual intervention.	
	a. Sessions should seamlessly failover to the next- superior link available without terminating the active sessions in case the performance of superior link gets degraded, without packet duplication. After restoration of the link, traffic should be moved back as per the last known good status. The seamless failover functionality shall work without traffic duplication on primary and secondary lines.	
	b. Should be able to maintain symmetric path for data traffic (send and receive the packets on same link for a session) at any point of time.	
High Availability		
46	The proposed SD-WAN Central Management solution/Controller/Orchestrator must be deployed in a High Availability at DC and DR	
47	The proposed SD-WAN Head-end Devices must be deployed in a High Availability at DC and DR	
48	The SD-WAN branch office devices should be deployed in standalone mode.	
Interoperability		
49	The proposed SDWAN solution should support static encrypted tunnel using well-known global standard protocol	
	1. any designated device of same OEM	
	2. any designated device of different OEM	

50	The devices in the proposed SDWAN solution should have interoperability for IPSec, routing protocol (BGP/static/OSPF), etc. with the existing products from multiple vendors in DOT's eco-system	
51	The proposed SDWAN solution should be able to integrate with AAA, AD, SIEM and NMS systems	
52	In the proposed SDWAN solution all the devices must support Port aggregation protocol with Open standards like LACP on LAN ports.	
53	In the Proposed SDWAN Interfaces of all devices (Central Management solution/Controller/Orchestrator/Head-end Devices/Branch Devices) must support virtual/sub interface (IEEE 802.1Q) on WAN and LAN side interface.	
Security		
54	All devices provisioned as part of the solution should have the capability to restrict management access based on IP address / Subnets.	
55	All the sensitive information (including passwords, keys) should be hashed (using SHA-2 or any other higher standards) and should not be visible in plain text at CLI / GUI.	
56	The encryption shall be done as per IPSec standards using AES-256 or AES-GCM or higher coupled with SHA 256/512 Hash Algorithm	
57	The proposed SDWAN devices/appliances should have authentication and authorization only with any of the pre-configured Central Management Solution/Controller/Orchestrator which are placed in Datacentres. Data should only flow after the device authentication and authorization is done by the Central Management Solution/Controller/Orchestrator.	
58	Proposed SDWAN devices should pull/push the configuration template from/to Central Management Solution/Controller/Orchestrator after authentication and authorization only.	
59	The solution should be able to upgrade edge/branch device from authenticated & authorized Central Management Solution/Controller/Orchestrator devices only.	
60	All devices access should be available through SSH (AES-128, AES-256 and higher, SHA-2 and higher) or HTTPS (TLS 1.2, TLS 1.3 or higher) and should support all future algorithms through upgrades or updates.	
61	SD-WAN must have geo-blocking system so that traffic can be block / filter based on locations	
62	SD-WAN Devices Local console access must have password-based authentication	
63	Logs should be configured to include but not limited to following: i. Authentication failures. ii. Account created/deleted/disabled. iii. Password changes for privileged accounts. iv. Changes in configuration settings. v. Start and stop of service. vi. System/Console alerts/errors or failures/fault logs. vii. Administrator or Root user activities. viii. Access to audit trails. ix. Creation and deletion of system-level objects. x. Alarms/alerts raised by the access control system (such as violation of any ACL defined)	

	xi. Details of system/files accessed of the SDWAN device. xii. Use of privileges. (i.e., Privilege escalation)	
64	All remote site traffic shall be encrypted when transported over WAN transport links: MPLS, ILL, FTTH, Broadband, SIM, ADSL/DSL, Ethernet, fibre, LTE/3G/4G/5G, satellite network protecting Data Confidentiality and Integrity.	
65	The proposed Head-end and Branch office SD-WAN devices shall have security functionalities like Firewall, Web Filtering and Intrusion Prevention.	
Routing and VPN		
66	The system should ensure that any change in physical connectivity (Link 1 to Link 2 connectivity in case of multiple links being terminated on the branch device) or any change in physical connectivity type (Link 1 connectivity changed from internet broadband to MPLS or vice versa, in case of multiple WAN links being terminated on the Branch device) does not require any change in virtual private network configuration in the Central Management Solution/Controller/Orchestrator or SD-WAN Devices.	
67	The system should support encapsulation types, but not limited to following: A. IPSEC B. GRE	
68	The system should support the following minimum authentication algorithms for Data Integrity: a. SHA-512 b. SHA-384 c. SHA-256	
69	The system should support the following minimum encryption algorithms for Data Security: a. AES-128 b. AES-256	
70	The system should allow time- based/scheduled manner refresh of the encryption key / standardized encryption algorithm for each virtual private network.	
71	The system should allow for alternate hub destinations to be created for application specific traffic using a policy defined for it.	
72	The proposed SD-WAN solution shall be able to auto configure VPN tunnels without limitation of the number of remote sites. The solution shall be able to auto configure VPN tunnels of a minimum of 2000 sites with and without manual VPN tunnel configurations.	
73	In case of a change in WAN IP Address (Private/Public) the SD-WAN solution shall detect and rebuild the VPN tunnel without manual VPN configurations.	
74	The proposed SD-WAN Solution shall be capable to view the routing table of the entire enterprise and have ability to audit and troubleshoot	
75	The tunnel creation should be automatic without any manual configuration on the edges and the controller.	
76	The proposed Solution should support route manipulation on LAN, WAN and Overlay to achieve various hybrid network integration requirements.	
77	The proposed Solution should support Site to Site IPSec tunnel on LAN/WAN side with any third party IPSEC capable appliance for hybrid deployments.	
78	The proposed Solution be capable of client-based VPN services	

NAT		
79	The SD-WAN solution shall support NAT and port forwarding of all LAN side prefixes.	
80	The SD-WAN solution shall support source NAT on location subnets to allow overlapping location subnet to communicate to the Datacentre. Also, the solution shall support LAN side destination NAT for simplicity of redirecting traffic	
QoS		
81	The solution shall be capable of configuring QoS weights on default applications, bespoke applications & destination IP based QoS definitions/VLAN based/VRF based etc	
82	The SD-WAN solution shall be able to apply QoS policies for all traffic types including TCP, UDP and other non-TCP traffic types	
83	The SD-WAN solution should include a QoS mechanism that is able to protect delay sensitive flows like Voice, Video.	
84	The device should support industry standard IP QoS mechanisms including (Traffic Class, IP Precedence and DSCP - Differentiated Services Code Point).	
85	QoS should work for all kind of traffic in both forward and backward direction. All the WAN ports of device should support QoS.	
86	The device should support (but not limited to) industry standard congestion management techniques such as class-based weighted fair queue, Low latency queue or equivalent.	
87	The device should support the per class bandwidth definition for QoS policy. The definition should support all the formats such as Kbps/Mbps/Gbps and percentage bandwidth. However, during configuration any one of these can be selected.	
88	The device should support separate QoS implementation (but not limited to) per wan link, per application (custom / defined), per class.	
89	The proposed SDWAN solution should be able to define minimum guaranteed bandwidth per class of service.	
WAN Optimisation		
90	Solution should have capability for WAN Optimization with following features	
	1. TCP Optimization	
	2. Sliding Window Adjustments on dynamic real time basis	
	3. UDP Optimization	
	4. Caching	
	5. Compression	
	6. Deduplication	
	7. Forward Equivalence Class	
	8. No traffic duplication for seamless failover of traffic	
91	The Solution shall have capability of Multi line WAN Optimization which means the WAN Optimization functions shall seamlessly continue on secondary line in case of primary link failure.	
92	The Solution shall have capability of Byte Level WAN Optimization for better granular performance.	

Centralized Management/Orchestrator/Controller		
93	The Centralized Management/Orchestrator/Controller shall provide a single, unified platform for network service provisioning, device configuration, monitoring and assurance and change management.	
94	The Centralized Management/Orchestrator/Controller should provide a dashboard that provides state of new appliances (Online, Offline, not connected).	
95	The Centralized Management/Orchestrator/Controller shall have web-based GUI. The solution shall support SSL for access to the management web GUI.	
96	The Centralized Management/Orchestrator/Controller should support a minimum of 2000 branch locations.	
97	The Centralized Management/Orchestrator/Controller should be able to segment the network with IP Subnets spanning across locations	
98	The Centralized Management/Orchestrator/Controller should be able to attach each local area network IP segment to a different network segment or by creating multiple virtual private networks.	
99	The Centralized Management/Orchestrator/Controller should have an option for integrated inbuilt DNS based feature for inbound load balancing of Internet and MPLS traffic and it shall be capable of Site Load Balancing in the same appliance to achieve DC/DR fail-over	
100	The Centralized Management/Orchestrator/Controller should have the capability to synchronize automatically as well as manually with geographically redundant devices. Automatic synchronization should be done in near real-time (within 60 seconds) without any manual intervention.	
101	The Centralized Management/Orchestrator/Controller should have the facility to configure, manage and upgrade the branch routers/devices centrally from orchestrator. The configuration on head-end / branch end must be minimal or equivalent to zero-touch provisioning. Configuration such as Interface IP address, orchestrator IP address to initiate initial reachability to Central Orchestrator, other parameter specific to each branch (ISP BGP neighbour IP, LAN IP etc.) can be locally done during the on boarding process of head-end/branch device. Rest all configurations to be done centrally from the orchestrator.	
102	The Centralized Management/Orchestrator/Controller should be capable of monitoring any configuration mismatch between orchestrator and branch device.	
103	The solution shall support end-to-end real-time flow visualization for the application paths for identifying issues and taking corrective actions.	
104	All network-wide configurations shall be from the centralized management/orchestrator/controller	
105	The Centralized Management/Orchestrator/Controller solution should support network wide device and network visibility for all the devices in the terminated on the devices irrespective of the type of link. Device should be equipped with the features like Visualize in real-time, Customized graphs and reports, Link Availability, WAN Link utilization, Detailed bandwidth usage of applications, Link wise WAN Link Latency, jitter and packet loss, SLA monitoring and compliance, Appliance load and	

	utilization, Alert for high resource utilization All types of alarms, Application performance monitoring.	
106	In case of performing any trouble shoot, the solution should have trouble tracking tools such as TCP dump, ping, trace route etc.	
107	The SD-WAN Solution shall support SYSLOG export directly from the SDWAN edge device	
	The solution shall support SNMP V2/V3, SYSLOG, IPFIX or equivalent polling of the SD-WAN devices and send SNMP to equivalent traps/logs. A dedicated SDWAN/Routing MIBs shall be available to poll SD-WAN specific functionality.	
	SD-WAN Central Management platform must be able to centrally manage software updates for SD-WAN appliance, ensuring they are up to date with the latest security patches and features	
Network Monitoring, Reports, Analytics and Notification		
108	The proposed solution to provide Traffic reports per site: availability, bandwidth usage per circuit (both upstream & downstream),latency, packet loss, QoS per link etc	
109	The proposed solution to provide Application reports: Applications usages, Bandwidth Consuming Applications.	
110	The solution must support device health monitoring for all the SD-WAN Devices in the network.	
111	Status of all the SD-WAN Devices deployed in WAN infrastructure should be viewed in a Graphical Map	
112	The solution must support events and email-based alarm to notify the administrators when any device/link fault or network performance degradation happens.	
113	The thresholds should be configurable per location /per SD-WAN Device.	
114	The alerts should be received for all WAN performance related events (like change in link characteristic, bandwidth, congestion)	
115	Edge device in the proposed solution should be able to send path parameters data, session details, etc. to the central reporting device at interval of every 5-minute maximum.	
116	The Dashboard should be able to display the current load on head-end and Branch devices	
117	The dashboard of the proposed solution should be capable of doing near real-time monitoring of all the links in GUI with respect to the following (but not limited to) parameters:	
	1. Packet loss	
	2. Jitter	
	3. Link errors	
	4. Bandwidth utilization	
	5. Latency	
	6. Duplex, Speed	
	The dashboard of the proposed solution should be capable of providing QoS Reports	
	The dashboard of the proposed solution should be capable of providing Application Reports	

118	SD-WAN Central Management Platform natively should be able to deliver monitor and provide report on the basis of following: 1. Device inventory 2. Session details 3. Changes in application usage and user activity 4. Users and applications that take up most of the network bandwidth 5. Uptime and downtime 6. Network activity 7. Tunnel activity	
119	SD-WAN Central Management Platform must deliver on-box downloadable reporting capability for application performance and link performance	
120	SD-WAN Central Management Platform should natively provide reports and same should be schedule for sent via email	
121	The proposed solution must be able to generate notifications for the following: 1. Link Flaps at the remote branch location 2. ISP link quality degrades 3. Link utilization along with threshold 4. CPU, Memory and Disk Utilization of the Branch Device	
122	The software defined network controller must have REST APIs available for 3rd party integration or integration with custom automation tools.	
RMA and Support		
123	The OEM should have a Technical Assistance Centre (TAC) available 24*7 with regional support centres for quick response to incidents.	
124	The proposed solution should have Direct Support from OEM (24*7) TAC Support and NBD replacement for the fault devices.	

Technical Specifications – Centralized Network Orchestrator/Manager/Controller		
Sr. No	**Requirements**	**Compliance Yes /No**
1	The proposed Centralized Network Orchestrator /Manager/Controller devices should have following minimum specifications	
	a. Minimum 4 Gbps throughput	
	b. Multicore CPU Architecture with minimum 8 Core processor.	
	c. Hardened 64-bit operating system	
	d. Minimum of 32 GB of RAM	
	e. Minimum 512 GB storage	
	f. Minimum interfaces from day 1: 4* 1G/10G Copper 4* 1G/10G Fiber	
2	In case of loss on connectivity with the SD-WAN Central Management Solution/Controller/Orchestrator, the SD-WAN Head-End and branch devices should continue to use the existing policy to forward the traffic. The data plane should not be affected even for single minute, with the loss of connectivity with the controller and Data plane traffic shall work without any interruption.	
3	The proposed SD-WAN Centralized Network Orchestrator /Manager/Controller devices must be capable of handling IPv6 on inward-facing interfaces and ISP outward-facing interfaces.	

4	The proposed SD-WAN Centralized Network Orchestrator /Manager/Controller devices shall be able to support Management of 500 devices from single Hardware.	
5	The proposed SD-WAN Centralized Network Orchestrator /Manager/Controller shall support high availability Architecture	
6	The Proposed SD-WAN Centralized Network Orchestrator /Manager/Controller devices shall support Geographic redundancy (500 KM) with high availability Architecture at each of the locations	
7	The proposed Centralized Network Orchestrator /Manager/Controller devices be supplied with, Power Cable and Rack Mount kit.	
8	The proposed SD-WAN Centralized Network Orchestrator /Manager/Controller should have Dual Power Supply Units	

Technical Specifications – Head End Devices		
Sr. No	**Requirements**	**Compliance Yes /No**
1	The proposed Head-end devices shall be able to connect in SD-WAN Architecture and shall meet requirements as per specification mentioned under functional requirement.	
2	The proposed SD-WAN Head-end Devices in the Datacentres should be able to peer with the existing routers in the Datacentres and shall support static and dynamic routing protocols.	
3	The proposed SD-WAN DC Devices should be Rack-mountable Physical appliances and should have Control Plane separated from the Data Plane in the Device Architecture itself.	
4	The Proposed SD-WAN Head-end Devices should have following minimum specifications	
	a. Minimum 20 Gbps throughput after enabling Firewall and Intrusion Prevention functionalities.	
	b. Multicore CPU Architecture with minimum 16 Core processor.	
	c. Hardened 64-bit operating system	
	d. Minimum of 256 GB of RAM expandable to 1TB	
	e. Minimum 512 GB storage	
	f. Must support at least - 200000 concurrent sessions	
	g. Site to Site VPN: Minimum 2500	
	h. Minimum Interfaces from Day 1: 8* 1G/10 G Copper 4* 1G/10 G Fiber	
5	Solution should be able bundle links that allows to group multiple physical links (that different ISPs use to communicate with the same destination) into a virtual SD-WAN interface.	
6	The SDWAN Head-end device should be able to automatically load balance and fail over MPLS, Internet, Leased Lines etc without traffic duplication with seamless failover capabilities. The VoIP and Video calls call shall not disconnect on failure of primary links as the traffic shall be migrated from primary to other available links in seamless manner without traffic duplication.	

7	In case of loss on connectivity with the Central Management Solution/Controller/Orchestrator, the proposed SD-WAN Head-End device should continue to use the existing policy to forward the traffic. The data plane should not be affected with the loss of connectivity with the controller and Data plane traffic shall work without any interruption.	
8	The proposed Head-end devices must be capable of handling IPv6 on inward-facing interfaces and ISP outward-facing interfaces. Dual stack support required from Day1.	
9	The proposed Head-end devices shall support failover of less than 20 Milliseconds.	
10	The proposed Head-end devices should be supplied with Power Cable and Rack Mount kit.	
11	The proposed Head-end devices should have integrated inbuilt DNS based feature for inbound load balancing of Internet and MPLS traffic and it shall be capable of Site Load Balancing in the same appliance to achieve DC & DR fail-over	
12	The proposed Head-end devices should have capability for WAN Optimization with following features	
	1. TCP Optimization	
	2. Sliding Window Adjustments on dynamic real time basis	
	3. UDP Optimization	
	4. Caching	
	5. Compression	
	6. Deduplication	
	7. Forward Equivalence Class	
	8. No traffic duplication for seamless failover of traffic	
	9. Forward Error correction	
13	The proposed Head-end devices shall have capability of Multi line WAN Optimization which means the WAN Optimization functions shall seamlessly continue on secondary line in case of primary link failure.	
14	The proposed Head-end devices shall have capability of Byte Level WAN Optimization for better granular performance.	
15	The proposed Head-end devices should support NAT IPV4 to 6 and vice versa for 200000 concurrent sessions.	
16	The proposed Head-end devices should have Redundant Power Supply Units Hot Swappable.	

Technical Specifications **Branch Devices – Type 1: District Treasury Offices**		
Sr. No	**Requirements**	**Compliance Yes /No**
1	The proposed Branch devices shall be able to connect in SD-WAN Architecture and shall meet requirements as per specification mentioned under functional requirement.	
2	The proposed SD-WAN Branch Devices should be Rack-mountable Physical appliances.	
	a. Minimum 200 Mbps throughput after enabling Firewall and Intrusion Prevention functionalities.	

	b. Multicore CPU Architecture with minimum 8 Core processor.	
	c. Hardened 64-bit operating system	
	d. Minimum of 32 GB of RAM and the appliance	
	e. Minimum 64 GB storage	
	f. Minimum Interfaces which are as follows 2* 1G/10 G Copper 4* 1G/10 G Fiber	
	g. Must support at least - 100000 concurrent sessions	
	h. Site to Site VPN: Minimum 100	
3	The proposed SD-WAN Branch Devices should be able to bundle links that allows to group multiple physical links (that different ISPs use to communicate with the same destination) into a virtual SD-WAN interface	
4	The SDWAN Branch devices should be able to automatically load balance and fail over VoIP, VPN, and Endpoints across divergent media types of MPLS, Internet, Leased Lines etc without traffic duplication with seamless failover capabilities i.e. the VoIP and Video calls call shall not disconnect on failure of primary links as the traffic shall be migrated from primary to other available links in seamless manner without traffic duplication.	
5	It should be possible to configure the LAN side subnets at branch location SD-WAN devices centrally from the Central Management Solution/Controller/Orchestrator.	
6	In case of loss on connectivity with the Central Management Solution/Controller/Orchestrator, the SD-WAN Branch Office devices should continue to use the existing policy to forward the traffic. The data plane should not be affected with the loss of connectivity with the controller and Data plane traffic shall work without any interruption.	
7	The proposed SD-WAN Branch Devices must be capable of handling IPv6 on inward-facing interfaces and ISP outward-facing interfaces. Dual stack support required from day 1.	
8	The proposed SD-WAN Branch Devices shall support less than 20 Milliseconds failover of connected last mile.	
9	The proposed SD-WAN Branch Devices should be supplied with Power Cable and Rack Mount kit.	
10	The proposed SD-WAN Branch Devices should have minimum 1 USB Dongle	

Technical Specifications Branch Devices – Type 2: Sub Treasury Offices		
Sr. No	**Requirements**	**Compliance Yes /No**
1	The proposed Branch devices shall be able to connect in SD-WAN Architecture and shall meet requirements as per specification mentioned under functional requirement.	
2	The proposed SD-WAN Branch Devices should be Rack-mountable Physical appliances.	
	a. Minimum 100 Mbps throughput after enabling Firewall and Intrusion Prevention functionalities.	

	b. Multicore CPU Architecture with minimum 8 Core processor.	
	c. Hardened 64-bit operating system	
	d. Minimum of 16 GB of RAM and the appliance	
	e. Minimum 64GB storage	
	f. Minimum Interfaces which are as follows 6* 1G/10 G Copper	
	g. Must support at least - 50000 concurrent sessions	
	h. Site to Site VPN: Minimum 100	
3	The proposed SD-WAN Branch Devices should be able to bundle links that allows to group multiple physical links (that different ISPs use to communicate with the same destination) into a virtual SD-WAN interface	
4	The SDWAN Branch devices should be able to automatically load balance and fail over VoIP, VPN, and Endpoints across divergent media types of MPLS, Internet, Leased Lines etc without traffic duplication with seamless failover capabilities i.e. the VoIP and Video calls call shall not disconnect on failure of primary links as the traffic shall be migrated from primary to other available links in seamless manner without traffic duplication.	
5	It should be possible to configure the LAN side subnets at branch location SD-WAN devices centrally from the Central Management Solution/Controller/Orchestrator.	
6	In case of loss on connectivity with the Central Management Solution/Controller/Orchestrator, the SD-WAN Branch Office devices should continue to use the existing policy to forward the traffic. The data plane should not be affected with the loss of connectivity with the controller and Data plane traffic shall work without any interruption.	
7	The proposed SD-WAN Branch Devices must be capable of handling IPv6 on inward-facing interfaces and ISP outward-facing interfaces. Dual stack support required from day 1.	
8	The proposed SD-WAN Branch Devices shall support less than 20 Milliseconds failover of connected last mile.	
9	The proposed SD-WAN Branch Devices should be supplied with Power Cable and Rack Mount kit.	
10	The proposed SD-WAN Branch Devices should have minimum 1 USB Dongle	

Annexure 7 (C) - Technical Specifications Network Monitoring cum Helpdesk Platform

Note: Following Technical Specifications compliances to be submitted on OEM Letterhead

Technical Specifications - Network Monitoring and Helpdesk Platform		
Sr. No	**Requirements**	**Compliance Yes /No**
A	The proposed Infrastructure Management System Platform should be from single OEM and shall support following functionalities:	
	1. General Requirements	
	2. Network Fault Monitoring & Performance Management with Reporting	
	3. Network Traffic Analysis	
	4. Helpdesk System, Incident, Problem, and Knowledge Management	
	5. Log Management	
1	**General Requirements**	
	Refer "It" as "The Proposed Network Monitoring cum Helpdesk Platform " in the below mentioned points	
1.1	Supply, Install, Integrate, Operate & Maintain the offered Network Monitoring cum Helpdesk Platform for 5 years. It should provide a unified reporting dashboard, integrated topology and event view.	
1.2	It should have a secured single sign-on and unified console for all functions of components offered for seamless cross functional navigation & launch for single pane of glass visibility across multiple areas of monitoring & management.	
1.3	It should be accessible via a Web based GUI console/portal from intranet as well as from internet.	
1.4	It should not come with any hidden license restriction to access services/features of proposed solution i.e. all the mentioned services of solution should be available from Single OEM.	
	There should not be any restriction on user creation and concurrent user sessions in the solution for any module except mentioned explicitly.	
1.5	It should have dual-stack IP support (support both IPv4 and IPv6) and should be completely vendor-agnostic in nature to be able to monitor a multi-vendor environment	
1.6	It should have helpdesk licenses of at least 5 concurrent user sessions for moderation of tickets/service requests. Also, the solution must provide access to 15 Field Technicians to provide feedback/update on the ticket/service request assigned by helpdesk personal.	
1.7	It should have Push API's using Web Socket to push data to 3rd Party Application for Dashboard and Reports (to optimize bandwidth usage).	
1.8	It should have Node Tags for device grouping and resource/interface tagging for element grouping. Apart from Node Tags additionally system should have options to do device grouping based on default fields and customer fields	
1.9	It should have reporting function to allow for building various reports and track the performance of Infrastructure and operational views in real time.	

1.10	It should have web-based Reporting Tool/Layer across various functional modules (such as fault, performance, etc.) proposed Platform. The Reporting Tool/Layer should provide ability to create, edit, and distribute report templates as well as on demand reporting to quickly design, save & share ad-hoc queries. It should create topology maps containing network devices discovered while generating the reports.	
1.11	It should support creating views to quickly extract the necessary info about events based on different predefined filters or search conditions.	
1.12	It should support Keyboard shortcuts to navigate through different modules and perform faster actions	
1.13	It should provide option for the users to choose personalized GUI Colour theme (Multi-Colour, Dark/Light Theme),	
	Multiple Logo supports based on the user state, system should load their state default logo.	
1.14	It should have IT Asset Management (IT and NON IT) and IP-Address Management features. In future without having to do any additional installations the ITAM and IPAM features can be used by buying and enabling the license.	
1.15	It should have a bi-directional integrated Network Configuration and Change Management (NCCM) & Network Detection & Response (NDR) tool with option to use NDR & NCCM features in future easily by buying and enabling the license for it without having to do any additional installations. The integration should allow assets and topology to sync from the NMS module to the NDR & NCCM features for helping in Root-Cause-Analysis of faults.	
2	**Refer "It " as "Network Fault Monitoring & Performance Management with Reporting" in the below mentioned points**	
2.1	It must monitor the performance across heterogeneous networks. It should automatically discover different type of heterogeneous devices (all SNMP supported devices i.e. Router, Switches, SD-WAN) and map the connectivity between them with granular visibility up to individual ports level. The tool shall be able to assign different icons/ symbols to different type of discovered elements. It should show live interface connections between discovered network devices.	
2.2	It must monitor and manage the devices with following attributes	
	a. Availability	
	b. Uptime	
	c. Network Latency	
	d. Usage/Throughput, Bandwidth	
	e. Error and Failure which include packet losses, interface failures, network failures etc.	
2.3	It should have a graphical topological display of all discovered network devices in real time.	
	The new devices must get added as an incremental discovery/partial without causing additional overhead of scanning the entire network. Further, any changes to devices must be identified in near real time during discovery scans	
2.4	It should support various discovery protocols to perform automatic discovery of all L2, L3 Network devices across DOT and any further Network connectivity's planned in future	

	System should have option for multiple options for discovery including IP address based discovery, IP address range discovery, CSV based discovery for bulk discovery and it should allow options to add custom fields to support customer specific data to upload during discovery	
2.5	It shall be able to discover Ipv4 only, Ipv6 only as well as devices in dual stack. In case of dual stack devices, the system shall be able to discover and show both Ipv4 and Ipv6 IP addresses.	
2.6	The discovery must also support device redundancy discovery in case of virtual IP addresses using vendor specific protocols such as VRRP and HSRP.	
2.7	Discovery has to work intelligently by identifying the device in the network by the given IP range and categorize into network devices and servers with vendor and model details.	
2.8	Automatically learn devices that supports SNMP, HTTP, Ping, SMTP, POP3, WMI, JMX, SOAP, REST API,PDC, SSH and Telnet along with any required protocol to communicate to the devices.	
2.9	It must provide a detailed asset report, organized by vendor name, device type, listing all ports for all devices. The proposed Network Fault Management console must also provide asset inventory reports	
2.10	It must provide reports to identify unused/dormant/administratively down Network ports in order to facilitate capacity planning	
2.11	It should be an integrated solution comprising of Fault, Performance Management features from including device fault and performance, Interface Faults and Performance, Bandwidth Analysis, Response time Analysis. It should proactively analyse problems to improve network performance.	
2.12	It should have extensive reporting facility, providing the ability to format and present data in a graphical and tabular display.	
2.13	It should be able to automatically generate a notification in the event of a link failure to ensure proper handling of link related issues. It should have a provision for suppression of maintenance alarms during the maintenance period.	
2.14	SLA calculation / Isolation report should be made with the consideration of both the Primary and Secondary link together instead of individual link based. The downtime calculation will be measured when both the links are down for internal reporting and link based for ISP reporting. System should provide the flexible configuration in UI itself based on user needs	
2.15	It should provide a notification mechanism that allows administrator to define what notification channel to be used in different time of days, and able to trigger multiple notifications to alert multiple person and actions	
2.16	It should allow QoS monitoring of WAN links across multiple technologies like Cisco IPSLA, Juniper RPM, Huawei NQA etc. across multiple protocols like HTTP, TCP, FTP, DNS etc.	
2.17	QoS parameters should include link response time, link-level latency, link-level packet loss, link-level jitter, Round-Trip-Time etc.	
2.18	It must support out of the box trend reports on group of metrics or group of devices in a single report. This will help understand the performance of multiple devices against a KPI. It must support of the box capacity planning reports that assist in the analysis of capacity needs based on projected load.	

2.19	It should be MIB-2 compliant by default. Moreover, NMS, should have ability to load the vendor specific MIBs. Furthermore, NMS should have capability to use multiple MIBs at a time for getting desired information, without the use of Software coding. Data Collection should be possible on MIB2 Expressions using any OID (Object Identifier)	
2.20	It shall be possible to use a KPI at different network element levels (individual Network Device, Interface, Group of Network Devices, Links, etc.) and time dimensions out of the box modifying the KPI definition. The system must support complex KPIs that are constructed by weighted combination of other KPIs.	
2.21	It should be capable to automatically send the reports through e-mail to pre-defined user with pre-defined interval.	
2.22	It should have capability to exclude the planned-downtimes and others for SLA purpose or necessary customization should be done to achieve the requirement.	
2.23	The performance management system shall be able to collect and report data like: Packet delay and packet loss, User bandwidth usage rate, Network availability rate, CPU usage rate, Input/output traffic through physical ports, Input/output traffic through logical ports	
2.24	It should support global threshold, and it should have option to define individual resource/interface statistics level threshold	
2.25	It should have threshold & Alerts: define and store static & dynamic resource utilisation thresholds, define and maintain availability thresholds for individual components, services or applications, individual service actions / automated operations (performing a transaction, running an automated procedure etc.), support comparison of actual availability, actual performance with agreed levels and alerting on breaches	
2.26	I It should have configurable parameters like frequency, data duration, resolution duration, reset points.	
2.27	All thresholds should have set point, reset point, polarity, and set point message and reset point message for ease of use.	
2.28	It should support separate Rule Engine based alarms apart from the generic threshold.	
	a. Should have capability to configure Device Group based, Node Based, Resources/Interface based, and Aggregation link based.	
	b. On Selection of Nodes/Resources/Aggregation links it have flexibility to filter based on fields available in node information	
	c. Rules should have option to apply configuration on top of performance value or based on configured threshold alarms	
	d. Rules should have option configure the breach based on min, max and average values	
	e. Should have option to configure rules and repeat counters	
	f. Should have options to select custom alarm and clear alarm messages for individual configured rules	
	g. Should have option to send severity levels like error, warning and information	
	h. Notifications support based on configured rules	
2.29	It supports sending alert via E-mail, SMS, Execute Batch file, SNMP Trap, XML notification, Pop-up window and Audio alert	

2.30	It should have Capacity Reservations: It should allow management of resource allocations and reservations (for services, applications or other needs), identify resource shortages and provide information for further analysis or procurement	
2.31	It should have Event Record & Classification: possible to generate event for all the monitoring devices, tool be used to define thresholds to generate events, collect from 3rd party using REST API , on regular interval Polling API and collect events from 3rd party system, classify them , assign different levels of severity to events, attribute events to specialised practices such as performance and capacity management, availability management, information security management, allow the automatic recording of event messages reported by external tools, filter & differentiate between different types of events	
2.32	It shall monitor all traffic from all the interfaces of the network device. Provides traffic Utilization based on individual interface level, nodes level or based on the group by location, branch, and departments as an Avg, Min and Max bandwidth, utilization, throughput or any custom monitoring parameters.	
2.33	It should have capability to configure business, non-business hours or custom time polling. These configurations should be available for every device as well as every component in the device.	
2.34	It should have the provision to disable and enable the polling of specific type of devices	
2.35	It should have capability to configure the maintenance period for any device. When device is in maintenance period there is no polling done and the SLA clock on the device is stopped.	
2.36	It should automatically generate daily reports that provide a summary of the Network Infrastructure as well as custom Reports and that are automatically sent by email at a pre-defined schedule to any recipient or save into any specific folder or drive.	
2.37	It should support instant diagnosis of the node status through Ping, Telnet and SNMP walk	
2.38	It should support Real-Time report generation for checking continuous reachability of target device	
2.39	It should provide many different types of topology representation, To perform the following:	
	1. Display physical connections of the different devices being monitored in the system	
	2. Display flat maps of the entire network or networks in a single view	
	3. Display customer maps based on user configurations	
	4. Display maps based on geo locations	
2.40	It should automatically learn IP Networks and their segments, LANs, hosts, switches, SD-WAN, etc. and to establish the connections and to correlate	
2.41	It should provide provision to draw & map user specific network diagram. It should have Integrated Web based feature to build Network Diagram	
2.42	It should be a Drag & Drop based Network Diagram builder, Dynamically Upload Images, Customizable objects to support multiple vendors, capability to export maps in an XML format and upload to any other system.	
2.43	It should provide Panel View of devices	
	a. Panel view should look similar to the actual device front panel	

	b. System should automatically detect the device model display the right panel without any additional configuration	
	c. Panel should show all the monitored interface with status	
	d. Fan status with live fan icon and LED status for power	
2.44	System should support VM, Hypervisor and Cluster monitoring from different vendors like VM-Ware, Citrix, Nutanix, Linux etc.	
2.45	System licensing should be based only on Physical Hosts and not charge separately for individual guest VMs running on VM Hosts	
3	**Refer "It " as "Network Traffic Analysis" in the below mentioned points**	
3.1	It should be able to monitor network traffic by capturing flow data from network devices. It shall be able to capture, track & analyse traffic flowing over the network via different industry standard traffic capturing methodologies viz. Net Flow, jflow, sFlow, IPFIX etc.	
3.2	It should identify which users, applications, protocols, countries, AS numbers, top routers, and top interfaces are consuming the most bandwidth	
3.3	It should have capability to alternatively capture traffic data via packet capture.	
3.4	It should be able to receive flows from non-SNMP-enabled devices, like VMware vSwitch	
3.5	It should monitor Type of Service (ToS), Differentiated Services Code Point (DSCP), and Per-Hop Behaviour (PHB),BGP AS and NEXT HOP	
4	**Refer "It " as "Dashboard" in the below mentioned points**	
4.1	It should provide Flexible, role based dashboards and operational consoles for a common understanding of device status, risks and quality problems.	
	This should give the right experts actionable information so IT executives can focus operations priorities, project owners understand IT status, and operations staff always knows which components in the network domains are impacted and can proactively mitigate issue.	
5	**IT Service Management (Helpdesk System, Incident, Problem, and Knowledge Management)**	
5.1	Helpdesk should support following operations:	
	Log incidents /Auto logging /issues as service requests and provide a unique service request number.	
	Acknowledgement would be sent to the user along with the service ticket number through an email immediately on issue logging.	
	All issues logged would be assigned a severity level (L1/L2 or L3). Indicative severity level definitions shall be discussed and finalized in consultation with the DOT at the time of deployment.	
	The Helpdesk application would provide workflow and hierarchy through which each incident should move based on Incident severity, classification and owner.	
	The Helpdesk staff shall have provisions through the application for coordinating with concerned vendor in case issues are pertaining to any external entity product/support like:	
	a. OEM team	
	b. Network bandwidth Provider	
	c. End User Devices support provider	
	Trouble Ticketing	

	a. User Interface to log tickets and complete the process	
	b. Workflow for the Incident management process with escalation mechanism (escalation hierarchy) and approval processes	
	c. Defining SLA as per agreed solution for resolving tickets and supporting auto-escalation of the tickets based on priority and SLA	
	Reporting	
	a. Provide Real-time Dashboard for reports	
5.2	Help Desk Tool should provide REST APIs to integrate with IT Infrastructure Management, Configuration Management, Network Management, CRM, SIEM tools to automate Events to Ticket.	
	The proposed tool must provide Dynamic REST/WEB API integration without writing any code, just by configuration on GUI with various inbuilt Authentication method.	
	Should be able to do 2 Way Integration with 3rd party and should be able to exchange the data at any stage of the lifecycle of the Event, Incident, Problem, Change, Request, Task, Release etc.	
5.3	Helpdesk Tool must provide intelligent Email-to-Incident feature. Tool should merge all subsequent email communication for a particular email-to-incident ticket into the same ticket in the form of a message thread. Tool should be intelligent enough to understand email conversation chains for merging emails to a particular incident. Merging logic should be not only based on Ticket ID but also on email sender, cc responses to that email chain	
5.4	Tool should be able to provide real-time Email, SMS Notification alerts to notify respective users about any changes in ticket state and status. Tool should provide Email Communication Interface to allow technicians to send replies to customers / end users from the tool GUI and Record all the Email Communication in Chronological Order	
5.5	The tool should provide a self-service portal where End-users / requesters can log in and raise incidents	
5.6	The proposed tool must provide GUI interface for users, requesters, customers, support staff, 3rd party vendors, Field Engineers, Site Engineers, Supervisors, Managers, etc. with options to restrict amount of information that can be accessed by each role.	
5.7	The integrated ITSM module should have its own Android & IOS app	
5.8	Tool should have option to define the workflow for each service created in the Service Catalogue for each process (Incident, Problem, Request)	
5.9	End user should also be able to browse the service catalogue and should be able to raise request by providing the input to particular service request, Address etc.	
5.10	The proposed tool must provide an option for users to track their incidents status, check knowledge articles and chat with technicians on particular incidents	
5.11	Users must have an option to Log, Classify, Categorize & Prioritize Requests	
5.12	The proposed tool must record Problem Record Date and Time, Problem Source, Contact Detail, Symptoms and Status	
5.13	Tool should allow Problem records to be classified according to priority and category	

5.14	Tool should allow a problem record to be escalated based on pre-established rules with option to manually overridden conditions	
5.15	The proposed tool must have a powerful knowledge management functionality	
5.16	In FAQ/Solutions type of knowledge, system should allow to add multiple questions/multiple solutions with single knowledge article	
6	**Log Management**	
6.1	The proposed solution should have option to collect and store system logs from target devices including firewalls, routers, switches, WLC, servers, SD-WAN devices, applications & databases	
6.2	The proposed solution should have multiple filtering options for incoming system logs based on target device, log ID, severity, level, message, OS type, application / database etc	
6.3	The proposed solution should have an option to export specific SYSLOG messages to users via email / SMS	

Annexure 7 (D) - Technical Specifications of Desktop PC

Technical Requirement of Desktop			
Sl. No	Aspects	Details	Compliance Yes/No
1	Processor	Intel i7 or higher	
2	Processor Generation	12th Gen or higher	
3	Processor base frequency (in GHz)	2.1 or better	
4	Chipset	Intel series (with M.2 support)	
5	RAM	32 GB DDR4	
6	SSD Capacity	1 TB (SSD)	
7	Internal speaker	To be available	
8	I/O ports	Minimum 4 USB ports among which a minimum of 2 USB ports of Version 3.0 and higher, Audio ports (Audio out, headphone out & microphone in ports), and HDMI ports	
9	Ethernet ports	10/100/1000 on board Integrated Gigabit Port	
10	Input Devices	Keyboard & Mouse (Wired or Wireless) of the same make as CPU	
11	Monitor	23.8 inches or higher TFT/LED Colour Monitor of the same make as CPU	
12	OS Certification	Windows 11 Professional with genuine factory pre-loaded software, whose certification details should be available on the OEM website.	
13	Warranty	Minimum of 5 Years	

Annexure 7 (E) - Technical Specifications of Professional Display Monitor

Technical Requirement for Professional Display Monitor			
Sl. No	Aspects	Details	Compliance Yes/No
1	Screen Size	Minimum 75 inches (diagonal)	
2	Display Type	LED or OLED, Anti-Glare Display	
3	Resolution	4K UHD (3840 x 2160) or higher	
4	Aspect Ratio	16:9	
5	Brightness	Minimum 500 nits	
6	Contrast Ratio	Minimum 1200:1	
7	Refresh Rate	60Hz or higher	
8	Viewing Angle	178° horizontal and vertical	
9	Inputs/Outputs	HDMI (minimum 2 ports, version 2.0 or higher) Display Port (optional, version 1.4 or higher) RS232 or RJ45	
10	Audio	Built-in speakers (minimum 10W x 2)	
11	Connectivity	Wi-Fi 802.11ac Bluetooth 4.2 or higher Remote Management Capabilities	
12	Power Supply	100-240V AC, 50/60Hz	
13	Warranty	Minimum of 5 Years	
14	Accessories	Remote control, wall mount kits, and required cables.	

Annexure 8 - Escalation Matrix - Bidder & OEM

Note for Bidder and OEM:

- To be submitted along with Technical Bid on Bidder and OEM Letterheads

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: Escalation Matrix
Tender Reference No:

Sl. No	Level	Role/Designation	Contact Person	Contact details

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal]

Annexure 9- Agreement

Note to Bidder:

- On a Stamp Paper of Relevant Value to be submitted Bidder

THIS AGREEMENT made on the….....Day of….....2025 between ….....(Name and Address of Purchaser) represented by the Secretary (Hereinafter "the Purchaser") on the one part and ……..... (Name and Address of Supplier) ….....(Hereinafter "the Supplier") represented by …….. (Name of the Authorized Signatory and Designation), Aged……. years, residing at (Full Residential Address of the Signatory) on the other part: WHEREAS the Purchaser has invited tenders for the supply of (brief description of goods and services vide tender no …dated the supplier has submitted technical and price bids and also demonstrated the technical specifications / features / other quality requirements as contained in the tender document.
The Purchaser has finalized the tender (Tender No. ……..……..Dated on ……..……)in favour of the Supplier for the supply of the said goods and services for a total cost of Rs.……(Contract Price in Words and Figures) (Hereinafter "the Contract Price") and issued Letter of Intent/ Supply Order No. …….…… dated ……..……...

NOW THIS AGREEMENT WITNESSETH AS FOLLOWS:

- In this Agreement words and expressions shall have the same meanings as are respectively assigned to them in the tender document referred to.
- The following documents shall be deemed to form and be read and constructed as part of this Agreement, viz.:
  - All the documents signed, stamped and submitted by the bidder as part of technical bid and price bid.
  - Signed and stamped all pages of tender documents
  - The Schedule of Requirements.
  - The detailed Technical Specifications and other quality parameters.
  - The compliance sheet submitted by the bidder
  - The clarifications and amendments issued / received as part of the tender document
  - The General Conditions of Contract.
  - The Specific Conditions of Contract; and all other terms & conditions mentioned in the tender i) The Purchaser's Letter of Intent.

- In consideration of the payments to be made by the Purchaser to the Supplier as hereinafter mentioned, the Supplier hereby covenants with the Purchaser to supply, install and commission the Goods and Services and to remedy defects therein in conformity in all respects with the provisions of the Contract.

- I/We, as supplier/bidder, hereby unconditionally accept that in case of any Hardware, software and any service failure, we shall extend our unconditional support as per the service level agreement (SLA) mentioned in this tender during the warranty and 3 Year CAMC period. As per the agreed terms, we here by agree to submit the bank guarantee upon signing the contract/agreement. During the valid contract period, if we fail to meet the SLA / revoke the contract, we are aware that the amount of the performance security is liable to be forfeited.

- We are also confirming that any information other than the products/services supplied by us, any information regarding the DOT assets/services / Records, may not be shared with any person outside of the DOT without written authorisation from the DOT.

- The Purchaser hereby covenants to pay the Supplier in consideration of the provision of the Goods and Services and the remedying of defects therein, the Contract Price or such other sum as may become payable under the provisions of the Contract at the times and in the manner prescribed by the Contract.

Purchaser DOT **Supplier / Bidder**

Annexure 10- Bidder & OEM Purchase Order Details

Note to Bidder and OEM:

- Please refer Section No.2.1Prequalification Criteria of Bidder
- To be submitted on Bidder's and OEM's Letterhead

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: Purchase Order Details
Tender Reference No:

Sl. No.	Customer Name & Details	Date of the Purchase Order (SD-WAN/ Network Monitoring cum Helpdesk Platform)	No of Devices (for SD-WAN/ Network Monitoring cum Helpdesk Platform PO's)	No of Multisite locations (SD-WAN PO's)
1				
2				
3				
4				

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal]

Annexure 11- Un-priced Bill of Material

Note to Bidder and OEM:

- To be submitted on Bidder's Letterhead

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: Un-priced Bill of Material
Tender Reference No:

Sl. No.	SD-WAN Solution Item Description	Model	Qty	Unit of Measurement
A.	**Hardware**			
1	Centralized Network Orchestrator/Manager/Controller DC-2(HA) and DR-2(HA)		4	No
2	Head-End Devices DC-2(HA) and DR-2(HA)		4	No
3	Branch Devices Type1 (Including 2 spare devices)		25	No
4	Branch Devices Type2 (Including 5 spare devices)		210	No
B.	**License and Support**			
5	License and Support	NA	60	Months
6	Support cost for 6th, 7th & 8th Year	NA	36	Months
7	Training - 4 Days	NA	25	Members
C.	**On-Site Support**			
8	One On-Site Technical Engineer	NA	12	Months
D.	**Additional Costs (If Applicable)**			
9	Any other Charges			

Sl. No.	Network Monitoring and Helpdesk Platform Item Description	Qty	Unit of Measurement
A. License			
1	Network Devices Monitoring (Switches and SD-WAN Devices)	600	No's
2	Network Device Monitoring: Named Users/Technicians	3	Members
3	Helpdesk: Named Users/Technicians	3	Members
4	Helpdesk: Field Technician	5	Members
5	Helpdesk: End Users	3500	Members
6	SYSLOG Collection Log Management SD-WAN Devices	250	No's
B. Support			
7	Support	60	Months
8	Support cost for 6th, 7th & 8th Year	36	Months
9	Training - 4 Days	25	Members
C. On-Site Support			
10	One On-Site Technical Engineer	12	Months

	E. Additional Costs (If Applicable)		
11	Any other Charges		

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal]

Annexure 12- Declaration on End of Sale or End of Support

Note to OEM:

- To be submitted on OEM's Letterhead

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: End of Sale or End of Support
Tender Reference No:

We hereby confirm that the proposed [Product/Solution Name] is neither at its End of Life (EoL) nor its End of Support (EoS) stage and will remain supported by [OEM Name] for a minimum of 5 years from the date of this declaration.

Product Name/Model: [Product/Solution Name or SKU]
EoL/EoS Status: Not Applicable
Support Commitment: Full technical and lifecycle support guaranteed for the next 5 years (until [Date]).

This declaration aligns with [OEM Name]'s product lifecycle policy, ensuring continued availability of updates, security patches, and technical support.

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal]

Annexure 13- Price Breakup

Note to Bidder:

- To be submitted on Bidder's Letterhead

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,

Subject: Price Breakup
Tender Reference No:

Sl. No.	SD-WAN Solution Item Description	Model	Qty	Unit of Measurement	Total Price Inclusive of all Taxes
A.	**Hardware**				
1	Centralized Network Orchestrator/Manager/Controller DC-2(HA) and DR-2(HA)		4	No	
2	Head-End Devices DC-2(HA) and DR-2(HA)		4	No	
3	Branch Devices Type1 (Including 2 spare devices)		25	No	
4	Branch Devices Type2 (Including 5 spare devices)		210	No	
B.	**Support**				
5	Support (SW and HW)	NA	60	Months	
6	Support cost for 6th, 7th & 8th Year	NA	36	Months	
7	Training - 4 Days	NA	25	Members	
C.	**On-Site Support**				
8	One On-Site Technical Engineer	NA	12	Months	
D.	**Additional Costs (If Applicable)**				
9	Any other Charges				

Sl. No.	Network Monitoring and Helpdesk Platform Item Description	Qty	Unit of Measurement	Total Price Inclusive of all Taxes
A. License				
1	Network Devices Monitoring (Switches and SD-WAN Devices)	600	No's	
2	Network Device Monitoring: Named Users/Technicians	3	Members	
3	Helpdesk: Named Users/Technicians	3	Members	
4	Helpdesk: Field Technician	5	Members	
5	Helpdesk: End Users	3500	Members	
6	SYSLOG Collection Log Management SD-WAN Devices	250	No's	
B. Support				

7	Support	60	Months	
8	Support cost for 6th, 7th & 8th Year	36	Months	
9	Training - 4 Days	25	Members	
C. On-Site Support				
10	One On-Site Technical Engineer	12	Months	
E. Additional Costs (If Applicable)				
11	Any other Charges			

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal]

Annexure 14 - Self Declaration by OEM for QCBS – Quality points

Note to OEM:

- To be submitted on OEM's Letterhead

To,
The Director, Directorate of Treasuries, Pattom, Pattom Palace, P.O. 695004 Trivandrum, Kerala

Dear Sir / Madam,
Subject: Self Declaration for QCBS - Quality points
Tender Reference No:

We, <The OEM>, hereby confirm to the below Quality points mentioned in the QCBS.

SL. No	Category	Description	Compliance Yes/No	Document to be submitted
1	SDWAN OEM Experience Government Department/BFSI/ PSU/PSE/Private sector	SDWAN Experience in India (As on Bid calling Date) a) More than 10 years - 5 Marks b) More than 7.5 years - 3 Marks c) More than 5 years - 2 Marks		PO copy with Date clearly mentioned
2		SDWAN Experience in India a) More than 500 location Project - 5 Marks b) More than 250 multisite Project - 3 Marks c) Less than 100 multisite Project - 2 Marks		PO copies
3	SD-WAN Technical Specifications (Core functionalities)	Capability to integrate different types of connectivity (MPLS/ILL/FTTH/Broadband/SIM/VSAT) into a single, end-to-end IP Framework and IP tunnel that uses policies to manage traffic.		Data Sheet and PoC
4		Capability to integrate different types of connectivity for Active- Active Load Balancing without traffic duplication on the connected Links at a site (MPLS/ILL/FTTH/Broadband/SIM/VSAT)		Data Sheet and PoC
5		Capability for DNS based Inbound Load Balancing of Multiple divergent media		Data Sheet and PoC
6		Capability of Seamless Failover of Voice/Video and Data Traffic less than 20Milliseconds		Data Sheet and PoC
7		Capability for switching traffic to new DHCP IP's from Service Providers for multiple DHCP based Links at sites		Data Sheet and PoC
8	SD-WAN Technical Specifications (Security)	Capability of inbuilt Firewall and IPS in DC/DR/Branch Devices		Data Sheet and PoC
9		Capability of Selective Encryption of Links at sites		Data Sheet and PoC
10		Traffic Encryption capability shall be FIPS 140-2 Compliant		Data Sheet and PoC
11	SD-WAN Technical Specifications	Capability to Prioritise and steer traffic based on Applications		Data Sheet and PoC

	(Application Awareness)			
12	SD-WAN Technical Specifications (WAN Optimization)	Built In Advanced WAN Optimisation Capabilities - Caching, Compression, Deduplication of Application traffic		Data Sheet and PoC
13		Capability of WAN Optimisation at Byte Level		Data Sheet and PoC
14	The proposed Network Monitoring and Helpdesk Platform OEM Experience	Network Monitoring cum Helpdesk Platform Experience in India - (in state/central Government/ PSU) a) Minimum 3 Projects with 10000 Devices - 5 Marks b) 2 Projects with 10000 Devices - 3 Marks c) 1 Project with 10000 Devices - 1 Marks		Reference PO copy and completion/ Signoff Documents need to be submitted.
15	The proposed Network Monitoring and Helpdesk Platform OEM certifications	a) The Proposed Network Monitoring cum Helpdesk Platform OEM should be OWASP & SANS certified from cert-in empanelled vendor. b) The Proposed Network Monitoring cum Helpdesk Platform OEM should be certified for ITIL V4 for 9 or more practices. c) The proposed Network Monitoring cum Helpdesk Platform. OEM should have ISO 270001, ISO 27034, SOC3 and CMMI Level 3		Documents need to be submitted for all 3 points
16	Presentation to Committee	The OEM SD-WAN and Network Monitoring cum Helpdesk Platform OEM to present (ON-LINE) the technical capabilities, features, proposed components, and deployment approach in line with RFP requirements.		Presentation to Technical Committee

- We (The OEM) agree to perform PoC on any parameter (or) all parameters mentioned in the QCBS Quality points by DoT at its discretion.
- We (The OEM) self-declare that "If we as an OEM, unable to perform the PoC for a particular criterion, where we have declared as complied for the mentioned in the QCBS Quality points – in which case, all the bid(s) of the respective OEM shall be disqualified / rejected from further evaluation, and we shall be liable for blacklisting.

Date:
Place:

[Signature]
(Name of Authorised Signatory)
Designation
[Company Seal]